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SPARSE POLYNOMIAL MATRIX OPTIMIZATION*

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Abstract. A polynomial matrix inequality (PMI) is a formula asserting that a polynomial matrix is positive semidefinite. Polynomial matrix optimization (PMO) concerns minimizing the smallest eigenvalue of a symmetric polynomial matrix subject to a tuple of PMIs. This work explores the use of sparsity methods in reducing the complexity of sum of squares–based methods in verifying PMIs or solving PMO. In the unconstrained setting, Newton polytopes can be employed to sparsify the monomial basis, resulting in smaller semidefinite programs. In the general setting, we show how to exploit different types of sparsity (term sparsity, correlative sparsity, matrix sparsity) encoded in polynomial matrices to derive sparse semidefinite programming relaxations for PMO. For term sparsity, we show that the block structures of the term sparsity iterations with maximal chordal extensions converge to the one determined by PMI sign symmetries. For correlative sparsity, unlike the scalar case, we provide a counterexample showing that asymptotic convergence does not hold under the Archimedean condition and the running intersection property. By employing the theory of matrix-valued measures, we establish several results on detecting global optimality and retrieving optimal solutions under correlative sparsity. The effectiveness of sparsity methods on reducing computational complexity is demonstrated on various examples of PMO.

Key words. polynomial matrix optimization, polynomial matrix inequality, moment-SOS hierarchy, term sparsity, correlative sparsity, semidefinite relaxation

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1. Introduction.

This paper is concerned with improving the tractability of verifying of polynomial matrix inequalities (PMIs) and of solving polynomial matrix optimization (PMO) problems. A PMI is a formula asserting that a polynomially defined symmetric matrix is positive semidefinite (PSD) over some given set [38]. PMIs are a generalization of (scalar) polynomial inequalities. Linear matrix inequalities are specific instances of PMIs restricted to degree one [19], but PMIs may describe nonconvex sets. Applications of PMIs include finding stability regions of autoregressive linear systems [21], sizing beams to minimize the cost of frame topologies under structural constraints [47], performing frequency-domain system identification of low-order linear systems under stability constraints [1], and simplifying optimal control tasks in the case where the applied input is constrained to lie in a semidefinite-representable set [35].

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Polynomial optimization problems (POPs) are programs where the objective and all constraints are defined by polynomials (forming a basic semialgebraic set). All POPs can be translated into equivalent programs with scalar linear objectives for which the constraint sets are described by polynomial inequality constraints. Checking nonnegativity of a polynomial over a basic semialgebraic set is generically an NP-hard problem, which extends (by reformulations in terms of membership oracles) to a generic NP-hardness of solving POPs [37]. Methods to verify polynomial nonnegativity include satisfiability solvers (e.g., dReal [11]) and sum of squares (SOS) methods [6].

If a real-valued polynomial with n variables can be represented as an SOS of other real-valued polynomials, then the original polynomial is certifiably nonnegative over the space $\mathbb{R}^n$. The set of SOS polynomials over $\mathbb{R}^n$ is strictly contained in the set of nonnegative polynomials over $\mathbb{R}^n$ [5]. Checking whether a polynomial is an SOS over $\mathbb{R}^n$ can be accomplished by solving a finite-dimensional convex optimization problem, which can be numerically treated using semidefinite programs (SDPs) [6]. SOS methods can also be used in the constrained setting to verify nonnegativity over basic semialgebraic sets. If the constraint set is compact and its representing constraints satisfy a real-algebraic Archimedean structure, then every positive polynomial over the basic semialgebraic set can have its positivity verified using SOS methods [43] (the polynomial degree needed to perform this verification is polynomial in the degree d for fixed n [4]).

These SOS methods for verifying polynomial nonnegativity have been extended to matrix SOS for PMIs [22], Hermitian SOS for complex polynomials [51], sums of Hermitian squares for noncommutative polynomials [18], and flag SOS for graph density polynomials [44]. The computational complexity of SOS implementation using SDPs is determined by the size and multiplicity of the largest PSD matrix constraint [29]. In the context of a $p \times p$ PMI with n variables where all polynomials are restricted to degree $2d$, the size of the largest PSD matrix (under the dense monomial basis) is $p\binom{n+d}{d}$. This matrix size (and the corresponding time to solve SDPs) suffers in a jointly polynomial manner as p, n, and d grow. Polynomial nonnegativity over sets admitting tractable Fourier analyses (e.g., ball, hypercube) can be accomplished in an optimization-free manner by checking a possibly exponential number of linear inequalities [10].

Existing knowledge of the problem structure can be used to reduce the computational expenditure of SOS-based verification [34, 56]. This decrease in complexity can be accomplished by a combination of reducing the size of the monomial basis used in formulating the polynomial representation and identifying opportunities for decomposing PSD matrix constraints in the SOS representation. Three dominant (and interlinked) approaches include correlative sparsity, term sparsity, and symmetry. Correlative sparsity structure ties together variables that appear in the same constraint or the same monomial term in the candidate polynomial [48]. Term sparsity structure pays attention to the monomials that appear in polynomials (commutative [52, 53], noncommutative [50]). The Newton polytope method in the unconstrained setting uses the exponents of monomials found in the candidate polynomial to determine which monomials can categorically be ruled out of an SOS representation [45]. In the context of symmetry, enforcing that an SOS polynomial is additionally invariant/equivariant with respect to a given group action adds severe structural constraints to its SOS representation. The resultant SOS decomposition involves a block-diagonal PSD structure where the polynomial basis is composed of primary and secondary invariants [12]. The term sparsity scheme in [53] can be interpreted as a refinement

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of symmetry reduction with respect to the class of sign symmetries [31]. Algebraic structure can be exploited if POPs are posed over regions described by polynomial equality constraints (in addition to polynomial inequality constraints), either by using Gröbner basis reduction over a quotient ring [42] or through the ideal-sparsity method by unfolding the separable equality constraints (e.g., $x(x-1)=0$). Multiple kinds of structures can be combined, such as the CS-TSSOS framework for correlative and term sparsity structures [54]. Term sparsity methods have also recently been used in sum-of-rational function optimization [16]. Term sparsity methods can also be used in the derivation of non-SOS certificates of polynomial nonnegativity. The sum of nonnegative circuit (SONC) framework decomposes a polynomial as the sum of "circuit" polynomials that are verifiably nonnegative due to their coefficients' satisfaction of the arithmetic mean-geometric mean (AM/GM) inequality [24]. The support and choice of these circuit polynomials are based on vertices of the Newton polytope (in the unconstrained setting). SONC verification therefore scales well when the candidate polynomial is sparse. SONC nonnegativity can be verified using relative entropy programs [8] or second-order cone programs [33, 49].

A basic PMO problem is of the form

$$\inf_{\boldsymbol{x}\in\mathbb{R}^n} \lambda_{\min}(F(\boldsymbol{x})) \quad \text{s.t.} \quad G_1(\boldsymbol{x}) \succeq 0, \ldots, G_m(\boldsymbol{x}) \succeq 0, \tag{1.1}$$

where $F, G_1, \ldots, G_m$ are symmetric polynomial matrices and $\lambda_{\min}(F(\boldsymbol{x}))$ denotes the smallest eigenvalue of $F(\boldsymbol{x})$. When F is a scalar polynomial (i.e., a 1×1 polynomial matrix), problem (1.1) is also known as a polynomial SDP which was extensively studied in the literature [20, 23, 26]. When both F and $G_1, \ldots, G_m$ are polynomial matrices, a matrix Moment-SOS hierarchy for solving (1.1) was provided in [15]. SOS-simplifying structure can be extended to the matrix case. When F is a polynomial matrix and $G_1, \ldots, G_m$ are polynomials, the work in [55] utilizes the matrix (chordal) sparsity of F to construct a sparse SOS representation for F. Note that matrix sparsity is independent of the presence of specific monomials in the nonzero entries. When F is a scalar polynomial, correlative sparsity was studied in [27, 39] and constraint matrix sparsity was considered in [25]. More recently, the work in [17] extends the term sparsity method to the case of scalar F in the context of frame topology optimization. We summarize different types of PMI sparsity in Figure 1. General sparsity methods for problem (1.1) have not previously been considered in the literature. This work fills this gap and studies sparsity reduction methods for PMO in full generality where F and $G_1, \ldots, G_m$ are all polynomial matrices.

This work makes the following contributions:

1. In the unconstrained setting, we propose a method based on Newton polytopes for reducing monomial bases.
2. We provide an iterative procedure to exploit term sparsity, which yields a bilevel hierarchy of sparse Moment-SOS relaxations for problem (1.1).

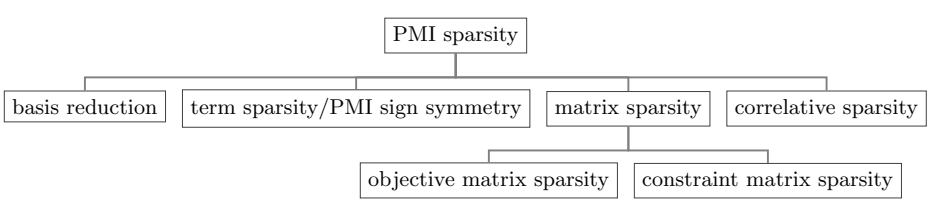

FIG. 1. *Different types of PMI sparsity.*

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In addition, we generalize the notion of (usual) sign symmetries for polynomials to the notion of PMI sign symmetries for polynomial matrices. It turns out that for PMO, the block structures of the term sparsity iterations with maximal chordal extensions converge to the one determined by PMI sign symmetries of problem (1.1).

3. We show how to exploit correlative sparsity for (1.1). When F is a scalar polynomial, it is known that asymptotic convergence of the correlatively sparse relaxations holds under the Archimedean condition and the running intersection property. However, when F is a polynomial matrix, we give a counterexample showing that asymptotic convergence does not hold under similar conditions. Moreover, by employing the theory of matrix-valued measures, we establish several results on detecting global optimality and retrieving optimal solutions in the correlatively sparse setting.
4. Decomposition methods based on the matrix sparsity structure of the objective matrix (extending the work of [55]) or PMI constraints are also provided.
5. Extensive numerical experiments are performed, which demonstrate the efficacy of our methods.

The rest of this paper is organized as follows. Section 2 introduces preliminaries such as notation, matrix SOS, matrix-valued measures, and graph theory. Section 3 applies the Newton polytope and term sparsity methods to unconstrained PMIs. Section 4 concerns the term sparsity method for constrained PMO. Section 5 investigates correlative sparsity for PMO. Section 6 considers matrix sparsity methods for PMO. Section 7 demonstrates the effectiveness of these methods via various numerical examples. Section 8 concludes the paper.

2. Preliminaries.

2.1. Notation. The n-dimensional real Euclidean vector space is $\mathbb{R}^n$. The set of natural numbers is $\mathbb{N}$, and the n-dimensional set of multi-indices is $\mathbb{N}^n$. The symbol $[n]$ denotes the set $\{1,\ldots,n\}$. For a set $\mathcal{A}$, its cardinality is denoted by $|\mathcal{A}|$. The degree of a multi-index $\boldsymbol{\alpha}\in\mathbb{N}^n$ is $|\boldsymbol{\alpha}| := \max_{i\in[n]}\alpha_i$. The associated monomial to an n-dimensional indeterminate $\boldsymbol{x} := (x_1,\ldots,x_n)$ and a multi-index $\boldsymbol{\alpha}\in\mathbb{N}^n$ is $\boldsymbol{x}^{\boldsymbol{\alpha}} = \prod_{i=1}^n x_i^{\alpha_i}$. Letting $\mathcal{A}\subset\mathbb{N}^n$ be a finite-cardinality set of multi-indices and $\{c_{\boldsymbol{\alpha}}\}_{\boldsymbol{\alpha}\in\mathcal{A}}$ be an associated set of real numbers, the polynomial formed by $\mathcal{A}$ and $\{c_{\boldsymbol{\alpha}}\}_{\boldsymbol{\alpha}\in\mathcal{A}}$ is $f(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}\in\mathcal{A}} c_{\boldsymbol{\alpha}}\boldsymbol{x}^{\boldsymbol{\alpha}}$. The set of polynomials with real-valued coefficients is $\mathbb{R}[\boldsymbol{x}]$. The *support* of a polynomial $f\in\mathbb{R}[\boldsymbol{x}]$, denoted by $\operatorname{supp}(f)$, is the set of multi-indices $\boldsymbol{\alpha}$ such that $c_{\boldsymbol{\alpha}}\neq 0$. The degree of a polynomial $f\in\mathbb{R}[\boldsymbol{x}]$ is $\deg f = \max_{\boldsymbol{\alpha}\in\operatorname{supp}(f)}|\boldsymbol{\alpha}|$. The set of polynomials of degree at most d is denoted by $\mathbb{R}[\boldsymbol{x}]_d$. Let $\circ$ denote the Hadamard (entrywise) product.

The transpose of a matrix A is denoted by $A^{\intercal}$. The p-dimensional identity matrix is I_p. The Kronecker product between two matrices C and D is $C\otimes D$. The set of $p\times p$ symmetric and real-valued matrices is $\mathbb{S}^p$. The subset of $p\times p$ PSD matrices is $\mathbb{S}^p_+$. Membership in the PSD set $Z\in\mathbb{S}^p_+$ will also be denoted as $Z\succeq 0$, and the Loewner partial ordering will be used as $Z_1\succeq Z_2 \Leftrightarrow Z_1 - Z_2\succeq 0$.

The set of polynomial matrices of dimension $p\times q$ is $\mathbb{R}[\boldsymbol{x}]^{p\times q}$. Given a polynomial matrix $Z(\boldsymbol{x}) = [Z(\boldsymbol{x})_{ij}]\in\mathbb{R}[\boldsymbol{x}]^{p\times q}$, we can write it as $Z(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}} Z_{\boldsymbol{\alpha}}\boldsymbol{x}^{\boldsymbol{\alpha}}$, $Z_{\boldsymbol{\alpha}}\in\mathbb{R}^{p\times q}$. The *support* of $Z(\boldsymbol{x})$, denoted by $\operatorname{supp}(Z)$, is the set of multi-indices $\boldsymbol{\alpha}$ such that $Z_{\boldsymbol{\alpha}}\neq 0$. The degree of $Z(\boldsymbol{x})$ is $\deg Z = \max_{i\in[p],\ j\in[q]}\deg Z(\boldsymbol{x})_{ij}$. The set of symmetric polynomial matrices of dimension p is $\mathbb{S}^p[\boldsymbol{x}]$. The set of p-dimensional symmetric polynomial matrices of degree at most d is denoted by $\mathbb{S}^p[\boldsymbol{x}]_d$.

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2.2. Sum of squares polynomials. This subsection will review SOS methods for verifying nonnegativity of polynomials over $\mathbb{R}^n$ and over semialgebraic sets.

2.2.1. Unconstrained. A polynomial $f(\boldsymbol{x}) \in \mathbb{R}[\boldsymbol{x}]_d$ is nonnegative over $\mathbb{R}^n$ if $f(\boldsymbol{x}) \geq 0$ for any $\boldsymbol{x} \in \mathbb{R}^n$. One method to verify polynomial nonnegativity is to use SOS certificates. A polynomial $f(\boldsymbol{x}) \in \mathbb{R}[\boldsymbol{x}]$ is SOS if there exist polynomials $g_1, \ldots, g_s \in \mathbb{R}[\boldsymbol{x}]$ such that $f(\boldsymbol{x}) = \sum_{i=1}^{s} g_i(\boldsymbol{x})^2$. The set of SOS polynomials, denoted by $\Sigma[\boldsymbol{x}]$, is contained inside the set of nonnegative polynomials over $\mathbb{R}^n$.

A polynomial matrix $Z(\boldsymbol{x}) \in \mathbb{S}^p[\boldsymbol{x}]$ is positive semidefinite (PSD) over $\mathbb{R}^n$ if $Z(\boldsymbol{x}) \succeq 0$ for all $\boldsymbol{x} \in \mathbb{R}^n$. Furthermore, we call $Z(\boldsymbol{x})$ an SOS matrix if there exists another polynomial matrix $R(\boldsymbol{x}) \in \mathbb{R}[\boldsymbol{x}]^{s\times p}$ such that $Z(\boldsymbol{x}) = R(\boldsymbol{x})^\intercal R(\boldsymbol{x})$. The set of n-dimensional SOS matrices is denoted by $\Sigma^n[\boldsymbol{x}]$. Note that the set of SOS (scalar) polynomials $\Sigma[\boldsymbol{x}]$ is equivalent to the set $\Sigma^1[\boldsymbol{x}]$ (the $n = 1$ case of SOS matrices).

Checking whether a polynomial matrix $Z(\boldsymbol{x}) \in \mathbb{S}^p[\boldsymbol{x}]_{2d}$ is inside the set $\Sigma^p[\boldsymbol{x}]$ can be accomplished by solving an SDP. Letting $\mathbf{m}_d(\boldsymbol{x})$ be the $\binom{n+d}{d}$-vector of monomials in $\boldsymbol{x}$ up to degree d, the matrix $Z(\boldsymbol{x})$ is an SOS matrix if there exists a PSD matrix $Q \in \mathbb{S}_+^{p\binom{n+d}{d}}$ such that (Lemma 1 of [46])

$$Z(\boldsymbol{x}) = (\mathbf{m}_d(\boldsymbol{x}) \otimes I_p)^\intercal Q(\mathbf{m}_d(\boldsymbol{x}) \otimes I_p). \tag{2.1}$$

Problem (2.1) is an SDP with respect to the PSD matrix constraint $Q \in \mathbb{S}_+^{p\binom{n+d}{d}}$ and the affine equality (coefficient matching) constraints in (2.1). Note that (2.1) can also be written as

$$Z(\boldsymbol{x}) = (I_p \otimes \mathbf{m}_d(\boldsymbol{x}))^\intercal \tilde{Q}(I_p \otimes \mathbf{m}_d(\boldsymbol{x})), \tag{2.2}$$

where $\tilde{Q}$ is obtained from Q by certain row and column permutations.

Verification of the PMI can also be accomplished through *scalarization.* The scalarization approach introduces a new tuple of variables $\boldsymbol{y} \in \mathbb{R}^p$ to form the equivalent problem:

$$\forall (\boldsymbol{x}, \boldsymbol{y}) \in \mathbb{R}^{n+p}: \qquad \boldsymbol{y}^\intercal Z(\boldsymbol{x})\boldsymbol{y} \geq 0. \tag{2.3}$$

The SOS-restriction of the scalarized problem (2.3) is

$$\boldsymbol{y}^\intercal Z(\boldsymbol{x})\boldsymbol{y} \in \Sigma[\boldsymbol{x}, \boldsymbol{y}]. \tag{2.4}$$

2.2.2. Constrained. Let $\mathbf{G} := \{G_k(\boldsymbol{x})\}_{k=1}^m$ be a set of symmetric polynomial matrices with $G_k(\boldsymbol{x}) \in \mathbb{S}^{q_k}[\boldsymbol{x}]$ which defines a basic semialgebraic set as

$$\mathbf{K} := \{\boldsymbol{x} \in \mathbb{R}^n \mid G_k(\boldsymbol{x}) \succeq 0 \quad \forall k \in [m]\}. \tag{2.5}$$

A polynomial matrix $Z(\boldsymbol{x}) \in \mathbb{S}^p[\boldsymbol{x}]$ satisfies a PMI over the constrained region $\mathbf{K}$ if

$$\forall \boldsymbol{x} \in \mathbf{K}: \qquad Z(\boldsymbol{x}) \succeq 0. \tag{2.6}$$

The following exposition of SOS verification of PMIs originates from [46].

Let C be a $pq \times pq$ block matrix described as

$$C = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1p} \\ C_{21} & C_{22} & \cdots & C_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ C_{p1} & C_{p2} & \cdots & C_{pp} \end{bmatrix} \tag{2.7}$$

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in which each block C_{ij} is of size $q \times q$. The p-product between C and a $q \times q$ matrix D is defined by

$$\langle C, D\rangle_p := \begin{bmatrix} \mathrm{tr}(C_{11}^\intercal D) & \mathrm{tr}(C_{12}^\intercal D) & \dots & \mathrm{tr}(C_{1p}^\intercal D) \\ \mathrm{tr}(C_{21}^\intercal D) & \mathrm{tr}(C_{22}^\intercal D) & \dots & \mathrm{tr}(C_{2p}^\intercal D) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{tr}(C_{p1}^\intercal D) & \mathrm{tr}(C_{p2}^\intercal D) & \dots & \mathrm{tr}(C_{pp}^\intercal D) \end{bmatrix}. \tag{2.8}$$

A sufficient SOS condition for (2.6) to hold is that there exist SOS matrices $S_0 \in \Sigma^p[\boldsymbol{x}], S_k \in \Sigma^{pq_k}[\boldsymbol{x}], k \in [m]$, such that

$$Z(\boldsymbol{x}) = S_0(\boldsymbol{x}) + \sum_{k=1}^{m} \langle S_k(\boldsymbol{x}), G_k(\boldsymbol{x})\rangle_p. \tag{2.9}$$

The dimension-p set of weighted sum of squares (WSOS) polynomial matrices $\Sigma^p[\mathbf{G}]$ is the subset of $\mathbb{S}^p[\boldsymbol{x}]$ that have a representation in (2.9) (also referred to the quadratic module formed by $\mathbf{G}$). The degree-$\leq 2d$ set of WSOS polynomials $\Sigma^p[\mathbf{G}]_{2d}$ is the subset of $\Sigma^p[\mathbf{G}]$ where $\deg S_0 \leq 2d$ and $\deg\langle S_k(\boldsymbol{x}), G_k(\boldsymbol{x})\rangle_p \leq 2d$ for all $k \in [m]$.

The quadratic module $\Sigma^p[\mathbf{G}]$ formed by $\mathbf{G}$ satisfies the *Archimedean* property if there exists an $a \geq 0$ such that the scalar polynomial $a - \|\boldsymbol{x}\|_2^2$ is a member of $\Sigma^1[\mathbf{G}]$. If $\Sigma^p[\mathbf{G}]$ is Archimedean, then $\mathbf{K}$ is compact, but not necessarily vice versa [9]. If there exists an $\epsilon > 0$ such that $Z(\boldsymbol{x}) \succeq \epsilon I_p$ over $\mathbf{K}$ and $\Sigma^p[\mathbf{G}]$ is Archimedean, then Scherer and Hol's Positivestellensatz [46, Corollary 1] states that $Z(\boldsymbol{x})$ will have a representation in (2.9). However, the degrees of the S terms needed to form the (2.9) certification may be exponential in n and $\deg Z$ (even for the $p = 1$ case) [40].

2.3. Matrix-valued measures and moment matrices. We recall some basics about the theory of matrix-valued measures and the associated moment matrices, which will be used to establish conditions for detecting global optimality and extracting optimal solutions from the Moment-SOS relaxations of PMO problems.

For the set $\mathbf{K}$ in (2.5), let $B(\mathbf{K})$ be the smallest σ-algebra generated from the open subsets of $\mathbf{K}$, and let $\mathfrak{m}(\mathbf{K})$ be the set of all finite Borel measures on $\mathbf{K}$. A measure $\phi \in \mathfrak{m}(\mathbf{K})$ is *positive* if $\phi(\mathcal{A}) \geq 0$ for all $\mathcal{A} \in B(\mathbf{K})$. Denote by $\mathfrak{m}_+(\mathbf{K})$ the set of all finite positive Borel measures on $\mathbf{K}$. The support $\mathrm{supp}(\phi)$ of a Borel measure $\phi \in \mathfrak{m}(\mathbf{K})$ is the (unique) smallest closed set $\mathcal{A} \in B(\mathbf{K})$ such that $\phi(\mathbf{K} \setminus \mathcal{A}) = 0$.

Let $\phi_{ij} \in \mathfrak{m}(\mathbf{K})$, $i, j = 1, \dots, p$. The $p \times p$ matrix-valued measure Φ on $\mathbf{K}$ is defined as the matrix-valued function $\Phi\colon B(\mathbf{K}) \to \mathbb{R}^{p\times p}$ with

$$\Phi(\mathcal{A}) := [\phi_{ij}(\mathcal{A})] \in \mathbb{R}^{p\times p} \quad \forall \mathcal{A} \in B(\mathbf{K}).$$

If $\phi_{ij} = \phi_{ji}$ for all $i, j = 1, \dots, p$, we call Φ a symmetric matrix-valued measure. If $\boldsymbol{v}^\intercal \Phi(\mathcal{A})\boldsymbol{v} \geq 0$ holds for all $\mathcal{A} \in B(\mathbf{K})$ and for all column vectors $\boldsymbol{v} \in \mathbb{R}^p$, we call Φ a PSD matrix-valued measure. The set $\mathrm{supp}(\Phi) := \bigcup_{i,j=1}^p \mathrm{supp}(\phi_{ij})$ is called the support of Φ. We denote by $\mathfrak{M}^p(\mathbf{K})$ (resp., $\mathfrak{M}_+^p(\mathbf{K})$) the set of all $p \times p$ (resp., PSD) symmetric matrix-valued measures on $\mathbf{K}$. A *finitely atomic PSD* matrix-valued measure $\Phi \in \mathfrak{M}_+^p(\mathbf{K})$ is a matrix-valued measure of form $\Phi = \sum_{i=1}^t W_i \delta_{\boldsymbol{x}^{(i)}}$, where $W_i \in \mathbb{S}_+^p$, $\boldsymbol{x}^{(i)}$'s are distinct points in $\mathbf{K}$, and $\delta_{\boldsymbol{x}^{(i)}}$ denotes the Dirac measure centered at $\boldsymbol{x}^{(i)}$, $i = 1, \dots, t$.

For a given matrix-valued measure $\Phi = [\phi_{ij}] \in \mathfrak{M}_+^p(\mathbf{K})$, we define a linear functional $\mathscr{L}_\Phi\colon \mathbb{S}[\boldsymbol{x}]^p \to \mathbb{R}$ as

$$\mathscr{L}_\Phi(F) = \int_{\mathbf{K}} \mathrm{tr}\,(F(\boldsymbol{x})\mathrm{d}\Phi(\boldsymbol{x})) = \sum_{i,j} \int_{\mathbf{K}} F_{ij}(\boldsymbol{x})\mathrm{d}\phi_{ij}(\boldsymbol{x}) \quad \forall F(\boldsymbol{x}) \in \mathbb{S}[\boldsymbol{x}]^p. \tag{2.10}$$

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Let $\mathbf{S} = (S_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha}\in\mathbb{N}^n}$ be a multi-indexed sequence of symmetric matrices in $\mathbb{S}^p$. We define a linear functional $\mathscr{L}_{\mathbf{S}}\colon \mathbb{S}[\boldsymbol{x}]^p \to \mathbb{R}$ in the following way:

$$\mathscr{L}_{\mathbf{S}}(F) := \sum_{\boldsymbol{\alpha}\in\operatorname{supp}(F)} \operatorname{tr}(F_{\boldsymbol{\alpha}} S_{\boldsymbol{\alpha}}) \quad \forall F(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}\in\operatorname{supp}(F)} F_{\boldsymbol{\alpha}}\boldsymbol{x}^{\boldsymbol{\alpha}} \in \mathbb{S}[\boldsymbol{x}]^p.$$

We call $\mathscr{L}_{\mathbf{S}}$ the *Riesz functional* associated to the sequence $\mathbf{S}$. The sequence $\mathbf{S}$ is called a matrix-valued $\mathbf{K}$-moment sequence if there exists a matrix-valued measure $\Phi = [\phi_{ij}] \in \mathfrak{M}_+^p(\mathbf{K})$ such that

$$\operatorname{supp}(\Phi) \subseteq \mathbf{K} \quad \text{and} \quad S_{\boldsymbol{\alpha}} = \int_{\mathbf{K}} \boldsymbol{x}^{\boldsymbol{\alpha}} \mathrm{d}\Phi(\mathbf{K}) := \left[\int_{\mathbf{K}} \boldsymbol{x}^{\boldsymbol{\alpha}} \mathrm{d}\phi_{ij}(\boldsymbol{x})\right]_{i,j\in[p]} \quad \forall \boldsymbol{\alpha}\in\mathbb{N}^n. \tag{2.11}$$

The measure $\Phi \in \mathfrak{M}_+^p(\mathbf{K})$ satisfying (2.11) is called a *representing measure* of $\mathbf{S}$.

DEFINITION 2.1. *Given a sequence $\mathbf{S} = (S_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha}\in\mathbb{N}^n} \subseteq \mathbb{S}^p$, the associated* moment matrix $M(\mathbf{S})$ *is the block matrix whose block row and block column are indexed by $\mathbb{N}^n$ and whose $(\boldsymbol{\alpha},\boldsymbol{\beta})$th block entry is $S_{\boldsymbol{\alpha}+\boldsymbol{\beta}}$ for all $\boldsymbol{\alpha},\boldsymbol{\beta}\in\mathbb{N}^n$. For $G\in\mathbb{S}[\boldsymbol{x}]^q$, the* localizing matrix $M(G\mathbf{S})$ *associated to $\mathbf{S}$ and G is the block matrix whose block row and block column are indexed by $\mathbb{N}^n$ and whose $(\boldsymbol{\alpha},\boldsymbol{\beta})$th block entry is $\sum_{\boldsymbol{\gamma}\in\operatorname{supp}(G)} S_{\boldsymbol{\alpha}+\boldsymbol{\beta}+\boldsymbol{\gamma}} \otimes G_{\boldsymbol{\gamma}}$ for all $\boldsymbol{\alpha},\boldsymbol{\beta}\in\mathbb{N}^n$. For $d\in\mathbb{N}$, the dth order moment matrix $M_d(\mathbf{S})$ (resp., localizing matrix $M_d(G\mathbf{S})$) is the submatrix of $M(\mathbf{S})$ (resp., $M(G\mathbf{S})$) whose block row and block column are both indexed by $\mathbb{N}_d^n$.*

Let $d_k := \lceil \deg G_k/2 \rceil$ for $k\in[m]$ and $d_{\mathbf{K}} := \max\{d_1,\ldots,d_m\}$. As in the scalar case, the existence of a finitely atomic representing measure of a matrix-valued sequence can be guaranteed by the "flatness condition."

THEOREM 2.1 ([15, Theorem 5]). *Given a truncated sequence $\mathbf{S} = (S_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha}\in\mathbb{N}_{2r}^n} \subseteq \mathbb{S}^p$, $\mathbf{S}$ admits an atomic representing measure $\Phi = \sum_{i=1}^t W_i \delta_{\boldsymbol{x}^{(i)}}$ with $W_i \in \mathbb{S}_+^p$, $\boldsymbol{x}^{(i)} \in \mathbf{K}$, and $\sum_{i=1}^t \operatorname{rank}(W_i) = \operatorname{rank}(M_r(\mathbf{S}))$ if and only if $M_r(\mathbf{S}) \succeq 0$, $M_{r-d_k}(G\mathbf{S}) \succeq 0$ for $k\in[m]$, and $\operatorname{rank}(M_r(\mathbf{S})) = \operatorname{rank}(M_{r-d_{\mathbf{K}}}(\mathbf{S}))$.*

We refer the reader to [15] for more details on matrix-valued measures/moments.

2.4. Polynomial matrix optimization and the moment-SOS hierarchy. The SOS matrix can be used to solve PMO problems. Consider the problem of finding an $\boldsymbol{x}\in\mathbf{K}$ to minimize the smallest eigenvalue of $F(\boldsymbol{x})$:

$$\lambda^\star := \sup_{\lambda\in\mathbb{R}} \lambda \quad \text{s.t.} \quad \forall \boldsymbol{x}\in\mathbf{K}\colon\ F(\boldsymbol{x}) \succeq \lambda I_p. \tag{2.12}$$

The SOS-restriction of problem (2.12) is

$$\sup_{\lambda\in\mathbb{R}} \lambda \quad \text{s.t.} \quad F(\boldsymbol{x}) - \lambda I_p \in \Sigma^p[\mathbf{G}], \tag{2.13}$$

and its rth order ($r \geq r_{\min} := \max\{\lceil \deg F/2\rceil, d_1,\ldots,d_m\}$) SOS-restriction is

$$\lambda_r^* := \sup_{\lambda\in\mathbb{R}} \lambda \quad \text{s.t.} \quad F(\boldsymbol{x}) - \lambda I_p \in \Sigma^p[\mathbf{G}]_{2r}. \tag{2.14}$$

The dual program of (2.14) (by convex duality) is an optimization problem posed over matrix-valued sequences $\mathbf{S} = (S_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha}\in\mathbb{N}_{2r}^n} \subseteq \mathbb{S}^p$ [15]:

$$\lambda_r := \begin{cases} \inf\limits_{\mathbf{S}} & \mathscr{L}_{\mathbf{S}}(F) \\ \text{s.t.} & M_r(\mathbf{S}) \succeq 0, \\ & M_{r-d_k}(G_k\mathbf{S}) \succeq 0, \quad k\in[m], \\ & \mathscr{L}_{\mathbf{S}}(I_p) = 1. \end{cases} \tag{2.15}$$

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It is clear that $\lambda_r^* \leq \lambda_r \leq \lambda^\star$ for all $r \geq r_{\min}$. When $\Sigma^p[\mathbf{G}]$ is Archimedean, Scherer and Hol's Positivestellensatz implies that $\lim_{r\to\infty} \lambda_r^* = \lim_{r\to\infty} \lambda_r = \lambda^\star$. The relaxations (2.14) and (2.15) indexed by relaxation order r is called the matrix Moment-SOS hierarchy for problem (2.12).

2.5. Graph theory. The term-sparse, correlatively-sparse, and matrix-sparse decomposition methods for PMIs explored in this paper use graph theory to identify exploitable structure in the optimization problems. A graph $\mathcal{G}(\mathcal{V},\mathcal{E})$ is defined by a collection of nodes $\mathcal{V}$ and edges $\mathcal{E}$. This paper will consider *undirected* graphs with self-loops. A complete graph is a graph in which every node is connected to every other node. A *clique* of $\mathcal{G}$ is a subgraph of $\mathcal{G}$ that is isomorphic to a complete graph. A *maximal clique* of $\mathcal{G}$ is a clique that is not a subgraph of another clique of $\mathcal{G}$.

A *path* in $\mathcal{G}$ exists between nodes v_i and v_j if there exists a sequence of nodes $v_i, v_1, v_2, \ldots, v_q, v_j$ such that $\{v_i, v_1\} \in \mathcal{E}, \{v_1, v_2\} \in \mathcal{E}, \ldots, \{v_q, v_j\} \in \mathcal{E}$. The nodes v_i and v_j are thus path-connected. The *block closure* of a graph $\mathcal{G}$ is the unique supergraph $\mathcal{G}' \supseteq \mathcal{G}$ where an edge is added between every pair of nodes that are path-connected. The graph $\mathcal{G}'$ is therefore isomorphic to a direct sum of complete graphs.

A *cycle* in a graph is a path that starts and ends with the same node. A graph is *chordal* if all its cycles of length at least four have a chord (i.e., an edge $\{v_i, v_j\}$ that joins two nonconsecutive nodes in the cycle). A *chordal extension* of $\mathcal{G}$ is a supergraph $\mathcal{G}' \supseteq \mathcal{G}$ with $\mathcal{G}'$ being a chordal graph. Determining if a graph is chordal can be done in linear time, but finding a chordal extension with a minimal number of edges is NP-hard [36]. Chordal extensions are typically not unique, and the block closure is a specific instance of a chordal extension. To distinguish from the block closure, we refer to an ordinary chordal extension by a *chordal closure.*[1] The block closure is the largest chordal extension of $\mathcal{G}$ such that disconnected components of $\mathcal{G}$ remain disconnected in $\mathcal{G}'$.

2.6. Sparse semidefinite programming. Chordal sparsity can be used to reduce the computational complexity needed to solve large-scale semidefinite programming problems. Given an undirected graph $\mathcal{G}(\mathcal{V},\mathcal{E})$ with nodes $\mathcal{V} = \{1,\ldots,p\}$, we denote the set of sparse symmetric matrices by $\mathbb{S}^p(\mathcal{G},0)$, i.e.,

$$\mathbb{S}^p(\mathcal{G},0) := \{A \in \mathbb{S}^p \mid A_{ij} = A_{ji} = 0 \quad \text{if } i \neq j \text{ and } \{i,j\} \notin \mathcal{E}\},$$

and let $\Pi_{\mathcal{G}} : \mathbb{S}^p \to \mathbb{S}^p(\mathcal{G},0)$ be the projection defined by

$$[\Pi_{\mathcal{G}}(A)]_{ij} = \begin{cases} A_{ij} & \text{if } i = j \text{ or } \{i,j\} \in \mathcal{E}, \\ 0 & \text{otherwise.} \end{cases} \tag{2.16}$$

We define the cone of sparse PSD matrices as $\mathbb{S}^p_+(\mathcal{G},0) := \mathbb{S}^p_+ \cap \mathbb{S}^p(\mathcal{G},0)$, and the dual cone of completable PSD matrices is given by

$$\mathbb{S}^p_+(\mathcal{G},?) = \Pi_{\mathcal{G}}(\mathbb{S}^p_+) = \{\Pi_{\mathcal{G}}(A) \mid A \in \mathbb{S}^p_+\}.$$

For any maximal clique $\mathcal{C}_i$ of $\mathcal{G}(\mathcal{V},\mathcal{E})$, we define the matrix $E_{\mathcal{C}_i} \in \mathbb{R}^{|\mathcal{C}_i| \times p}$ by

$$[E_{\mathcal{C}_i}]_{jk} = \begin{cases} 1 & \text{if } \mathcal{C}_i(j) = k, \\ 0 & \text{otherwise,} \end{cases}$$

[1] In this paper, we implement the "MinimumDegree" greedy algorithm in [7] to generate a chordal closure.

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where $\mathcal{C}_i(j)$ denotes the jth node in $\mathcal{C}_i$, sorted in the natural ordering. Given $A \in \mathbb{S}^p$, the matrix $E_{\mathcal{C}_i}$ can be used to extract a principal submatrix indexed by $\mathcal{C}_i$, i.e., $E_{\mathcal{C}_i} A E_{\mathcal{C}_i}^\intercal \in \mathbb{S}^{|\mathcal{C}_i|}$. Given $A \in \mathbb{S}^{|\mathcal{C}_i|}$, the operation $E_{\mathcal{C}_i}^\intercal A E_{\mathcal{C}_i}$ inflates A into a sparse $p \times p$ matrix.

When the sparsity pattern graph $\mathcal{G}$ is chordal, the cone $\mathbb{S}_+^p(\mathcal{G}, 0)$ can be decomposed as a sum of simple convex cones, as stated in the following theorem.

THEOREM 2.2 ([2, Theorem 2.3]). *Let $\mathcal{G}(\mathcal{V}, \mathcal{E})$ be a chordal graph and $\{\mathcal{C}_1, \ldots, \mathcal{C}_t\}$ be the set of maximal cliques of $\mathcal{G}$. Then, $A \in \mathbb{S}_+^p(\mathcal{G}, 0)$ if and only if there exist matrices $A_i \in \mathbb{S}_+^{|\mathcal{C}_i|}$, $i = 1, \ldots, t$, such that $A = \sum_{i=1}^t E_{\mathcal{C}_i}^\intercal A_i E_{\mathcal{C}_i}$.*

By duality, Theorem 2.2 leads to the following characterization of the PSD completable cone $\mathbb{S}_+^p(\mathcal{G}, ?)$.

THEOREM 2.3 ([14, Theorem 7]). *Let $\mathcal{G}(\mathcal{V}, \mathcal{E})$ be a chordal graph, and let $\{\mathcal{C}_1, \ldots, \mathcal{C}_t\}$ be the set of its maximal cliques. Then, $A \in \mathbb{S}_+^p(\mathcal{G}, ?)$ if and only if $E_{\mathcal{C}_i} A E_{\mathcal{C}_i}^\intercal \in \mathbb{S}_+^{|\mathcal{C}_i|}$ for all $i \in [t]$.*

3. Term sparsity for unconstrained PMO. This section will detail term sparsity methods for unconstrained PMO, or equivalently for simplifying SOS verification of a single PMI $F(\boldsymbol{x}) \succeq 0$ in the fashion of the decomposition in (2.2). The first subsection will discuss how to reduce the required monomial basis, thus setting some diagonal elements of the Gram matrix Q to zero. The second subsection will define matrix term sparsity patterns to impose a block structure on the matrix Q.

3.1. Selecting the monomial basis. Letting $F(\boldsymbol{x}) \in \mathbb{S}^p[\boldsymbol{x}]_{2d}$ be a polynomial matrix, define $\{\boldsymbol{v}^j(\boldsymbol{x})\}_{j=1}^p$ as a set of monomial (column) vectors such that

$$F(\boldsymbol{x}) = \operatorname{diag}(\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x}))^\intercal \, Q \operatorname{diag}(\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x})), \tag{3.1}$$

where $\operatorname{diag}(\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x}))$ is the matrix formed by putting $\boldsymbol{v}^j(\boldsymbol{x})$ at the jth diagonal position with all other elements being zeros. The formulation in (2.2) is an instance where for all $j \in [p]$, $\boldsymbol{v}^j(\boldsymbol{x}) = \mathbf{m}_d(\boldsymbol{x})$ (the full monomial basis). The PMI $\forall \boldsymbol{x} \in \mathbb{R}^n : F(\boldsymbol{x}) \succeq 0$ is verified if the Gram matrix satisfies $Q \succeq 0$. This subsection will use the monomial structure of $F(\boldsymbol{x})$ to design bases $\{\boldsymbol{v}^j(\boldsymbol{x})\}$ that may have a lower cardinality than the full basis $\mathbf{m}_d(\boldsymbol{x})$. The basis reduction will be accomplished by using Newton polytope techniques on the scalarized PMI (2.4).

DEFINITION 3.1. *The Newton polytope of a polynomial $f \in \mathbb{R}[\boldsymbol{x}]$ is defined as* $\operatorname{New}(f) = \operatorname{conv}(\operatorname{supp}(f))$, *where* $\operatorname{conv}(\cdot)$ *means taking the convex hull.*

Newton polytopes can be used to reduce the number of active monomials in an SOS decomposition.

THEOREM 3.1 (Theorem 1 of [45]). *Let f be a polynomial such that $f \in \Sigma[\boldsymbol{x}]_{2d}$, and let $g \in \mathbb{R}[\boldsymbol{x}]^t$ be a vector of polynomials such that $f(\boldsymbol{x}) = \sum_{j=1}^t g_j(\boldsymbol{x})^2$. Then the support of each factor g_j is constrained by* $\operatorname{New}(g_j) \subseteq \frac{1}{2}\operatorname{New}(f)$.

A Gram SOS representation $f(\boldsymbol{x}) = \boldsymbol{v}(\boldsymbol{x})^\intercal Q \boldsymbol{v}(\boldsymbol{x})$ can therefore restrict the monomial vector $\boldsymbol{v}(\boldsymbol{x})$ to those corresponding to all integer points in $\frac{1}{2}\operatorname{New}(f)$, rather than choosing all $\binom{n+d}{d}$ monomial elements in the full basis $\mathbf{m}_d(\boldsymbol{x})$. The size of the Gram matrix Q can therefore be reduced (ensuring that solving the SDP is more efficient), at the preprocessing cost of finding all integer points in the 1/2-scaled Newton polytope.

We now return to the PMI setting with a polynomial matrix $F(\boldsymbol{x}) \in \mathbb{S}^p[\boldsymbol{x}]$. Monomial bases $\{\boldsymbol{v}^j(\boldsymbol{x})\}$ from (3.1) can be determined by virtue of Theorem 3.1 as follows.

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THEOREM 3.2. *Let $F(\boldsymbol{x}) \in \Sigma^p[\boldsymbol{x}]$ be a polynomial matrix and the exponent set $\mathcal{A}^j$ be chosen as $\mathcal{A}^j := \mathbb{N}^n \cap \frac{1}{2}\mathrm{New}(F(\boldsymbol{x})_{jj})$, where $F(\boldsymbol{x})_{jj}$ is the jth diagonal element of $F(\boldsymbol{x})$, $j \in [p]$. Then the monomial basis $\boldsymbol{v}^j(\boldsymbol{x})$ from* (3.1) *can be chosen without conservatism as $\boldsymbol{v}^j(\boldsymbol{x}) = \{\boldsymbol{x}^{\boldsymbol{\alpha}} \mid \boldsymbol{\alpha} \in \mathcal{A}^j\}$.*

Proof. Consider the scalarization $\boldsymbol{y}^\intercal F(\boldsymbol{x})\boldsymbol{y}$ of $F(\boldsymbol{x})$ from (2.3). Since $\boldsymbol{y}^\intercal F(\boldsymbol{x})\boldsymbol{y}$ is a quadratic form in $\boldsymbol{y}$, it holds that each integral point $(\boldsymbol{\alpha}_{\boldsymbol{x}}, \boldsymbol{\alpha}_{\boldsymbol{y}}) \in \mathbb{N}^{n+p} \cap \frac{1}{2}\mathrm{New}(\boldsymbol{y}^\intercal F(\boldsymbol{x})\boldsymbol{y})$ satisfies $|\boldsymbol{\alpha}_{\boldsymbol{y}}| = 1$. For each $j \in [p]$, let $\tilde{\mathcal{A}}^j := \{\boldsymbol{\alpha}_{\boldsymbol{x}} \in \mathbb{N}^n \mid (\boldsymbol{\alpha}_{\boldsymbol{x}}, \boldsymbol{\alpha}_{\boldsymbol{y}}) \in \mathbb{N}^{n+p} \cap \frac{1}{2}\mathrm{New}(\boldsymbol{y}^\intercal F(\boldsymbol{x})\boldsymbol{y}), \boldsymbol{\alpha}_{y_j} = 1\}$ correspond to the set of $\boldsymbol{x}$-monomials multiplied by y_j. By Theorem 3.1, one can choose $\boldsymbol{v}^j(\boldsymbol{x}) = \{\boldsymbol{x}^{\boldsymbol{\alpha}} \mid \boldsymbol{\alpha} \in \tilde{\mathcal{A}}^j\}$ without conservatism. Moreover, the sets satisfy $\tilde{\mathcal{A}}^j = \mathcal{A}^j$ by construction. ∎

The procedure in Theorem 3.2 will be outlined with an example.

Example 3.1. Let $F(\boldsymbol{x}) \in \mathbb{S}^3[\boldsymbol{x}]$ be defined by

$$F(\boldsymbol{x}) = \begin{bmatrix} 1 + x_1^4 - 0.5x_1^2 + 0.25x_1 + x_2^2 + x_3^4 & -x_2 & x_1 + x_3 \\ -x_2 & 2 + 2x_2^4 + 2x_1^2 - x_1 & x_1 \\ x_1 + x_3 & x_1 & 2 + 3x_3^2 + x_1^2x_2^2 \end{bmatrix}. \tag{3.2}$$

By respectively computing the Newton polytopes of the diagonal elements $F(\boldsymbol{x})_{11}$, $F(\boldsymbol{x})_{22}$, $F(\boldsymbol{x})_{33}$, the monomial bases $\{\boldsymbol{v}^j(\boldsymbol{x})\}$ can be given as

$$\boldsymbol{v}^1(\boldsymbol{x}) = \{1, x_1^2, x_1, x_1x_3, x_2, x_3^2, x_3\}, \tag{3.3a}$$

$$\boldsymbol{v}^2(\boldsymbol{x}) = \{1, x_1, x_2, x_2^2\}, \tag{3.3b}$$

$$\boldsymbol{v}^3(\boldsymbol{x}) = \{1, x_1x_2, x_3\}. \tag{3.3c}$$

There are 14 monomials in total listed in (3.3). Verification that $F(\boldsymbol{x})$ from (3.2) is an SOS matrix can therefore be accomplished by solving an SDP with a Gram matrix $Q \in \mathbb{S}_+^{14}$ rather than in $\mathbb{S}_+^{30}$ (as would be required with a full choice of all $\binom{3+2}{3} = 10$ monomials for each of the three columns). An admissible SOS decomposition $F(\boldsymbol{x}) = R(\boldsymbol{x})^\intercal R(\boldsymbol{x})$ found by the reduced bases in (3.3) is

$$R(\boldsymbol{x}) \approx \begin{bmatrix} 1 + 0.125x_1 - 0.517x_1^2 & 0.0864x_1 & x_3 \\ 0.0760x_1 + 0.8543x_1^2 & -0.617 + 0.0525x_1 & 0.608x_3 \\ 0.707x_1 & 0.707x_1 & 1.414 \\ x_2 & -1 & 0 \\ x_3^2 & 0 & 0 \\ -0.050x_1 & 0.787 - 0.594x_1 & 0.477x_3 \\ -0.016x_1 & 1.066x_1 & 0.155x_3 \\ 0 & 1.414x_2^2 & 0 \\ -0.1223x_1 & 0 & 1.174x_3 \\ 0 & 0 & x_1x_2 \end{bmatrix}. \tag{3.4}$$

3.2. Term sparsity patterns. The previous subsection concentrated on reducing the number of elements in the monomial bases $\{\boldsymbol{v}^j(\boldsymbol{x})\}$. This subsection will focus on reducing the maximal size of PSD matrices involved in the SDP through the application of term sparsity.

DEFINITION 3.2. *Let $\mathbf{B} := \{\boldsymbol{v}^1(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x})\}$ be the concatenation of the vectors $\boldsymbol{v}^j(\boldsymbol{x})$. The* term sparsity pattern (TSP) *of a polynomial matrix $F(\boldsymbol{x}) \in \mathbb{S}^p[\boldsymbol{x}]$ is an undirected graph including self-loops with nodes corresponding to entries of $\mathbf{B}$, and*

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edges are drawn between nodes $a \in \boldsymbol{v}^i(\boldsymbol{x})$ *and* $b \in \boldsymbol{v}^j(\boldsymbol{x})$ *if* $ab \in \{\boldsymbol{x}^{\boldsymbol{\alpha}} \mid \boldsymbol{\alpha} \in \operatorname{supp}(F_{ij})\}$ *or if* ab *is a monomial square given that* $i = j$.

Example 3.2. Consider the bivariate matrix

$$F(\boldsymbol{x}) = \begin{bmatrix} 1 + x_1^2 + 2x_1^2x_2^2 + x_2^2 & x_1x_2 \\ x_1x_2 & 2 + x_1^2x_2^2 + x_2^4 \end{bmatrix} \tag{3.5}$$

admitting an SOS-matrix decomposition $F(\boldsymbol{x}) = R(\boldsymbol{x})^\intercal R(\boldsymbol{x})$ with

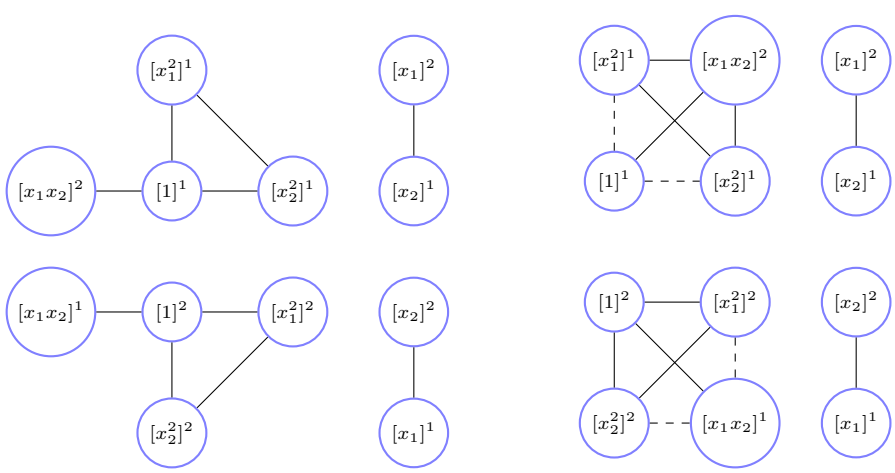
$$R(\boldsymbol{x}) = \begin{bmatrix} x_1 & x_2 \\ \sqrt{2}x_1x_2 & 0 \\ x_2 & 0 \\ 1 & 0 \\ 0 & \sqrt{3/2} \\ 0 & x_1x_2 \\ 0 & x_2^2/\sqrt{2} \\ 0 & (x_2^2 - 1)/\sqrt{2} \end{bmatrix}. \tag{3.6}$$

This example uses the standard full bases $\boldsymbol{v}^1(\boldsymbol{x}) = \boldsymbol{v}^2(\boldsymbol{x}) = \{1, x_1, x_2, x_1^2, x_1x_2, x_2^2\}$. The j-indices of a monomial $a \in \boldsymbol{v}^j(\boldsymbol{x})$ for $j \in \{1, 2\}$ will be indicated by a bracketed superscript as $[a]^j$. The TSP of (3.5) is drawn in Figure 2(a). This graph has five connected components. As an example, the off-diagonal term $F(\boldsymbol{x})_{12} = x_1x_2$ could be created by multiplications of $[1]^1, [x_1x_2]^2$ or $[x_1]^1, [x_2]^2$.

The term sparsity method proceeds by the iterative repetition of two operations on the TSP graph [53].

- *Support extension:* Add edges to the graph based on the sets of active monomials.
- *Chordal extension:* Add edges to the graph by performing a chordal extension.

More concretely, for each pair $i, j \in [p]$, define the initial support set associated to the (i, j)-position by



(a) chordal closure. (b) block closure.

FIG. 2. *TSP graph of* $F(\boldsymbol{x})$ *in* (3.5) *and its chordal/block closure.*

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$$\mathcal{C}_{i,j}^{(0)} := \begin{cases} \operatorname{supp}(F_{ii}) \cup \boldsymbol{v}^i(\boldsymbol{x})^{\circ 2} & \text{if } i = j, \\ \operatorname{supp}(F_{ij}) & \text{otherwise,} \end{cases} \tag{3.7}$$

where $\boldsymbol{v}^i(\boldsymbol{x})^{\circ 2}$ denotes the set of squares of monomials in $\boldsymbol{v}^i(\boldsymbol{x})$. For $s \geq 0$, the support extension operation is defined as

$$\text{SupportExtension}(\mathcal{E}^{(s)}) := \bigcup_{i,j=1}^{p} \left\{ \{[a]^i, [b]^j\} \in \boldsymbol{v}^i(\boldsymbol{x}) \times \boldsymbol{v}^j(\boldsymbol{x}) \,\middle|\, ab \in \mathcal{C}_{i,j}^{(s)} \right\}. \tag{3.8}$$

Next, we perform a chordal extension on the graph with $\text{SupportExtension}(\mathcal{E}^{(s)})$ as its edges,

$$\mathcal{E}^{(s+1)} \leftarrow \text{ChordalExtension}(\text{SupportExtension}(\mathcal{E}^{(s)})), \tag{3.9}$$

and update the support set

$$\mathcal{C}_{i,j}^{(s+1)} := \left\{ ab \,\middle|\, \{[a]^i, [b]^j\} \in \mathcal{E}^{(s+1)} \right\}. \tag{3.10}$$

There are two typical choices for the chordal extension operation in (3.9): an (approximately) smallest chordal closure (i.e., chordal extension with the smallest tree-width) and the block closure. The block closure would offer a block-diagonal structure on the Gram matrix, whereas the sparser chordal closure could potentially contain overlaps among maximal cliques. The chordal extension does not add any edges to the TSP graph in Figure 1. Figure 2 displays the block closure and highlights new edges as dashed lines.

The above process is guaranteed to stabilize after a finite number of iterations, because the cardinality of the node set $\mathbf{B}$ is finite and both the support extension and chordal extension steps only add edges. After obtaining the graph $\mathcal{G}^{(s)}(\mathbf{B}, \mathcal{E}^{(s)})$ at the sth term sparsity step ($s \geq 1$), we may consider the following sparse SDP to decompose $F(\boldsymbol{x})$ as an SOS:

$$\begin{cases} \text{Find} & Q \in \mathbb{S}_+^{|\mathbf{B}|}(\mathcal{G}^{(s)}, 0) \\ \text{s.t.} & F(\boldsymbol{x}) = \operatorname{diag}(\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x}))^\intercal \, Q \operatorname{diag}(\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x})). \end{cases} \tag{3.11}$$

THEOREM 3.3. *Assume that the chordal extension in* (3.9) *is chosen to be the block closure and the process in* (3.9) *stabilizes at the sth step. Then,* $F(\boldsymbol{x}) \in \Sigma^p[\boldsymbol{x}]$ *if and only if* (3.11) *is feasible.*

Proof. The proof is similar to the counterpart in [53, Theorem 3.3]. ∎

4. Term sparsity for constrained PMO. This section will apply term sparsity techniques to PMO constrained by PMIs. Specifically, we extend the iterative procedure on exploiting term sparsity for scalar constrained polynomial optimization [52, 53] to the situation of PMO. Similarly as in the unconstrained case of section 3.2, the term sparsity decomposition will proceed based on alternating support extension and chordal extension steps.

Consider the PMO problem

$$\lambda^\star := \inf_{\boldsymbol{x} \in \mathbb{R}^n} \lambda_{\min}(F(\boldsymbol{x})) \quad \text{s.t.} \quad G_1(\boldsymbol{x}) \succeq 0, \ldots, G_m(\boldsymbol{x}) \succeq 0, \tag{4.1}$$

where $F = [F_{ij}] \in \mathbb{S}^p[\boldsymbol{x}]$ and $G_k = [G_k^{i,j}] \in \mathbb{S}^{q_k}[\boldsymbol{x}], k \in [m]$. By abuse of notation, we denote a monomial basis $\{\boldsymbol{x}^{\boldsymbol{\alpha}}\}_{\boldsymbol{\alpha} \in \mathcal{B}}$ by its exponent set $\mathcal{B}$ in the following. For a

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monomial basis $\mathcal{B}$ and a positive integer k, we use $\mathcal{B}^{(k)}$ to denote the vector of k-copies of $\mathcal{B}$, that is,

$$\mathcal{B}^{(k)} := (\underbrace{\boldsymbol{\alpha}, \ldots, \boldsymbol{\alpha}}_{k\text{-copies}})_{\boldsymbol{\alpha} \in \mathcal{B}}. \tag{4.2}$$

For convenience, the k-copies of $\boldsymbol{\alpha} \in \mathcal{B}$ in $\mathcal{B}^{(k)}$ are respectively labeled $[\boldsymbol{\alpha}]^1, \ldots, [\boldsymbol{\alpha}]^k$.

Recall that $d_k := \lceil \deg G_k/2 \rceil$ for $k \in [m]$ and $r_{\min} := \max\{\lceil \deg F/2 \rceil, d_1, \ldots, d_m\}$. Fix a relaxation order $r \geq r_{\min}$. Set $d_0 := 0$ and $q_0 := 1$. Let

$$\mathbf{B}_{r,k} := (\mathbb{N}^n_{r-d_k})^{(pq_k)}, \quad k = 0, \ldots, m, \tag{4.3}$$

and for each pair $i, j \in [p]$, let us define the initial support set

$$\mathcal{C}^{(0)}_{i,j} := \begin{cases} \operatorname{supp}(F_{ii}) \cup (2\mathbb{N}^n_r) & \text{if } i = j, \\ \operatorname{supp}(F_{ij}) & \text{otherwise.} \end{cases} \tag{4.4}$$

Now for each $k \in [m]$, we iteratively define an s-indexed ascending chain of graphs $(\mathcal{G}^{(s)}_{r,k}(\mathbf{B}_{r,k}, \mathcal{E}^{(s)}_{r,k}))_{s \geq 1}$ via two successive operations:

(1) *Support extension.* Add edges to the graph $\mathcal{G}^{(s)}_{r,k}$ according to the sets of activated supports $\mathcal{C}^{(s)}_{i,j}$:

$$\operatorname{SupportExtension}(\mathcal{E}^{(s)}_{r,k}) := \bigcup_{i,j=1}^{pq_k} \Bigg\{ \{[\boldsymbol{\alpha}]^i, [\boldsymbol{\beta}]^j\} \in \mathbb{N}^n_{r-d_k} \times \mathbb{N}^n_{r-d_k} \;\Bigg|\; \Big(\boldsymbol{\alpha} + \boldsymbol{\beta} + \operatorname{supp}(G_k^{\bar{i},\bar{j}})\Big) \cap \mathcal{C}^{(s)}_{\lceil i/q_k \rceil, \lceil j/q_k \rceil} \neq \emptyset \Bigg\},$$

where for any $i \in [pq_k]$,

$$\bar{i} := \begin{cases} i \pmod{q_k} & \text{if } q_k \nmid i, \\ q_k & \text{otherwise.} \end{cases}$$

(2) *Chordal extension.* For each k, perform a chordal extension on the graph with $\operatorname{SupportExtension}(\mathcal{E}^{(s)}_{r,k})$ as its edge set, i.e., let

$$\mathcal{E}^{(s+1)}_{r,k} \leftarrow \operatorname{ChordalExtension}(\operatorname{SupportExtension}(\mathcal{E}^{(s)}_{r,k})), \quad k = 0, \ldots, m, \tag{4.5}$$

and for each $i, j \in [p]$, update the sets of activated supports:

$$\mathcal{C}^{(s+1)}_{i,j} := \bigcup_{k=0}^{m} \bigcup_{i',j'=1}^{q_k} \bigcup_{\{[\boldsymbol{\alpha}]^{(i-1)q_k+i'}, [\boldsymbol{\beta}]^{(j-1)q_k+j'}\} \in \mathcal{E}^{(s+1)}_{r,k}} \Big(\boldsymbol{\alpha} + \boldsymbol{\beta} + \operatorname{supp}(G_k^{i',j'})\Big). \tag{4.6}$$

Clearly, the above process is guaranteed to stabilize after a finite number of iterations for all k, because the cardinality of each node set $\mathbf{B}_{r,k}$ is finite and both the support extension and chordal extension steps only add edges.

Given an undirected graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, we use the symbol $B_{\mathcal{G}} \in \{0,1\}^{|\mathcal{V}| \times |\mathcal{V}|}$ to refer to the adjacency matrix of $\mathcal{G}$ whose diagonal are all ones. Then, for $s \geq 1$, the corresponding sparse moment relaxation for (4.1) is given by

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$$\lambda_r^{(s)} := \begin{cases} \inf & \mathscr{L}_{\mathbf{S}}(F) \\ \text{s.t.} & B_{\mathcal{G}_{r,0}^{(s)}} \circ M_r(\mathbf{S}) \in \mathbb{S}_+(\mathcal{G}_{r,0}^{(s)}, ?), \\ & B_{\mathcal{G}_{r,k}^{(s)}} \circ M_{r-d_k}(G_k\mathbf{S}) \in \mathbb{S}_+(\mathcal{G}_{r,k}^{(s)}, ?), \quad k \in [m], \\ & \mathscr{L}_{\mathbf{S}}(I_p) = 1. \end{cases} \tag{4.7}$$

We call s the *sparse order* associated with problem (4.7).

PROPOSITION 4.1. *The following statements are true:*

(i) *Fixing a relaxation order $r \geq r_{\min}$, the sequence $\{\lambda_r^{(s)}\}_{s\geq 1}$ is monotonically nondecreasing and $\lambda_r^{(s)} \leq \lambda^\star$ for all $s \geq 1$.*

(ii) *Fixing a sparse order $s \geq 1$, the sequence $\{\lambda_r^{(s)}\}_{r\geq r_{\min}}$ is monotonically nondecreasing.*

(iii) *Assume that the chordal extension in* (4.5) *is chosen to be the block closure. Then for any $r \geq r_{\min}$, the sequence $\{\lambda_r^{(s)}\}_{s\geq 1}$ converges to λ_r in finitely many steps.*

Proof. The proofs are similar to the counterparts in [52, 53]. ∎

Term sparse decompositions are intricately linked to *PMI sign symmetries* of the underlying polynomial matrices.

DEFINITION 4.1. *A* PMI sign symmetry *of a polynomial matrix $P(\boldsymbol{x}) \in \mathbb{S}^p[\boldsymbol{x}]$ is a binary vector $\theta \in \{-1,1\}^n$ such that either $P(\theta \circ \boldsymbol{x}) = P(\boldsymbol{x})$ or there exists a complete bipartite graph $\mathcal{G}^\theta(\mathcal{V},\mathcal{E})$ with $\mathcal{V} = [p]$ and satisfying*

(1) $[P(\theta \circ \boldsymbol{x})]_{ij} = [P(\boldsymbol{x})]_{ij}$ *if $i = j$ or $\{i,j\} \notin \mathcal{E}$;*

(2) $[P(\theta \circ \boldsymbol{x})]_{ij} = -[P(\boldsymbol{x})]_{ij}$ *if $\{i,j\} \in \mathcal{E}$.*

In addition, we define the binary matrix $E^{P,\theta} \in \{-1,1\}^{p\times p}$ such that $P(\theta \circ \boldsymbol{x}) = E^{P,\theta} \circ P(\boldsymbol{x})$.

The name "PMI sign symmetry" is justified by the following proposition.

PROPOSITION 4.2. *Let $P(\boldsymbol{x}) \in \mathbb{S}^p[\boldsymbol{x}]$, and let θ be a PMI sign symmetry of $P(\boldsymbol{x})$. If $P(\boldsymbol{x}) \succeq 0$, then $P(\theta \circ \boldsymbol{x}) \succeq 0$.*

Proof. Assume that $E^{P,\theta}$ is not an all-ones matrix. One can easily show that $E^{P,\theta} \succeq 0$ from the fact that $\mathcal{G}^\theta(\mathcal{V},\mathcal{E})$ is a complete bipartite graph. Therefore, we have $P(\theta \circ \boldsymbol{x}) = E^{P,\theta} \circ P(\boldsymbol{x}) \succeq 0$. ∎

The PMI sign symmetry is a generalization of the usual sign symmetry (corresponding to $P(\theta \circ \boldsymbol{x}) = P(\boldsymbol{x})$). The latter was studied in [30] to yield a block-diagonal structure for matrix SOS decompositions. It turns out that the more general PMI sign symmetry can also give rise to a block-diagonal structure for matrix SOS decompositions.

THEOREM 4.3. *Suppose that $F(\boldsymbol{x}) \in \mathbb{S}^p[\boldsymbol{x}]$ admits a representation:*

$$F(\boldsymbol{x}) = \operatorname{diag}(\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x}))^\intercal \, Q \operatorname{diag}(\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x})) \tag{4.8}$$

with $Q \succeq 0$. Let Θ collect all PMI sign symmetries of $F(\boldsymbol{x})$. We define a $\{0,1\}$-binary matrix B indexed by $\{\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x})\}$ via

$$B_{[a]^i,[b]^j} = \begin{cases} 1 & \text{if } \left([a]^i \cdot [b]^j\right)(\theta \circ \boldsymbol{x}) = E_{ij}^{F,\theta} \cdot \left([a]^i \cdot [b]^j\right)(\boldsymbol{x}) \quad \forall \theta \in \Theta, \\ 0 & \text{otherwise.} \end{cases} \tag{4.9}$$

Then, it holds that

$$F(\boldsymbol{x}) = \operatorname{diag}(\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x}))^\intercal \, (B \circ Q) \operatorname{diag}(\boldsymbol{v}^1(\boldsymbol{x}), \boldsymbol{v}^2(\boldsymbol{x}), \ldots, \boldsymbol{v}^p(\boldsymbol{x})). \tag{4.10}$$

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Proof. Let us consider the scalarization $\boldsymbol{y}^\intercal F(\boldsymbol{x})\boldsymbol{y}$ of $F(\boldsymbol{x})$. It can be seen that the PMI sign symmetries of $F(\boldsymbol{x})$ are in a one-to-one correspondence to the usual sign symmetries of $\boldsymbol{y}^\intercal F(\boldsymbol{x})\boldsymbol{y}$. Then, the block-diagonal structure in (4.10) follows from the block-diagonal structure given by the sign symmetries of $\boldsymbol{y}^\intercal F(\boldsymbol{x})\boldsymbol{y}$ [31]. ∎

Theorem 4.2 can be further extended to the constrained case as follows.

THEOREM 4.4. *Let $F \in \mathbb{S}^p[\boldsymbol{x}]$ and $G_k \in \mathbb{S}^{q_k}[\boldsymbol{x}], k \in [m]$. Suppose that F admits a representation:*

$$F = (I_p \otimes \mathbf{m}_r(\boldsymbol{x}))^\intercal Q_0 (I_p \otimes \mathbf{m}_r(\boldsymbol{x})) + \sum_{k=1}^m \langle (I_{pq_k} \otimes \mathbf{m}_{r-d_k}(\boldsymbol{x}))^\intercal Q_k (I_{pq_k} \otimes \mathbf{m}_{r-d_k}(\boldsymbol{x})), G_k \rangle_p,$$

where $Q_0, Q_1, \ldots, Q_m \succeq 0$. Let Θ collect the common PMI sign symmetries of $F, G_1, \ldots, G_m$. Let B_0 be defined in the same manner as B in Theorem 4.3. Moreover, for each $k \in [m]$, we define a $\{0,1\}$-binary matrix B_k indexed by the monomial basis $\{\mathbf{m}^1_{r-d_k}(\boldsymbol{x}), \ldots, \mathbf{m}^{pq_k}_{r-d_k}(\boldsymbol{x})\}$ via

$$[B_k]_{[a]^{(i-1)q_k+\ell},[b]^{(j-1)q_k+w}} = \begin{cases} 1 & \text{if } c(\theta \circ \boldsymbol{x}) = E^{F,\theta}_{ij} \cdot E^{G_k,\theta}_{\ell w} \cdot c(\boldsymbol{x}) \quad \forall \theta \in \Theta, \\ 0 & \text{otherwise,} \end{cases}$$

where $c := [a]^{(i-1)q_k+\ell} \cdot [b]^{(j-1)q_k+w}$. Then, F admits the following block-diagonal representation:

$$F = (I_p \otimes \mathbf{m}_r(\boldsymbol{x}))^\intercal (B_0 \circ Q_0) (I_p \otimes \mathbf{m}_r(\boldsymbol{x})) + \sum_{k=1}^m \langle (I_{pq_k} \otimes \mathbf{m}_{r-d_k}(\boldsymbol{x}))^\intercal (B_k \circ Q_k) (I_{pq_k} \otimes \mathbf{m}_{r-d_k}(\boldsymbol{x})), G_k \rangle_p. \tag{4.11}$$

The proof of Theorem 4.4 also utilizes the scalarization trick, but it is more technical and we omit it here for the sake of conciseness. As a corollary of Theorem 4.4, we immediately obtain the following PMI sign symmetry adapted Positivstellensatz for polynomial matrices.

COROLLARY 4.5. *Let $F \in \mathbb{S}^p[\boldsymbol{x}]$ and $G_k \in \mathbb{S}^{q_k}[\boldsymbol{x}], k \in [m]$. Suppose that F is positive definite on $\mathbf{K}$ defined in* (2.5) *and the related quadratic module $\Sigma^p[\mathbf{G}]$ satisfies the Archimedean property. Let $\{B_k\}_{k=0}^m$ be the $\{0,1\}$-binary matrices provided in Theorem 4.4 by considering the common PMI sign symmetries of $F, G_1, \ldots, G_m$. Then, F admits a block-diagonal representation* (4.11) *for some $Q_0, Q_1, \ldots, Q_m \succeq 0$.*

Proof. It follows from Scherer and Hol's Positivestellensatz and Theorem 4.4. ∎

For scalar polynomial optimization, the block structures of the term sparsity iterations with block closures converge to the one determined by sign symmetries [53]. This result could be extended to the case of PMO by considering PMI sign symmetries.

THEOREM 4.6. *Consider the PMO problem* (4.1). *The block structures of the term sparsity iterations with block closures converge to the one determined by the common PMI sign symmetries of $F(\boldsymbol{x}), G_1(\boldsymbol{x}), \ldots, G_m(\boldsymbol{x})$.*

Proof. We assume that $q_1 = \cdots = q_m = 1$. The proof for the general case is more involved but follows in a similar manner. Now the PMO problem (4.1) is equivalent to the following scalar polynomial optimization problem:

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$$\inf_{\boldsymbol{x}\in\mathbb{R}^n,\boldsymbol{y}\in\mathbb{R}^p} \boldsymbol{y}^\intercal F(\boldsymbol{x})\boldsymbol{y} \quad \text{s.t. } G_1(\boldsymbol{x})\geq 0,\ldots,G_m(\boldsymbol{x})\geq 0,\ 1-y_1^2-\cdots-y_p^2=0. \tag{4.12}$$

Comparing the term sparsity iterations for (4.1) described in this work with the ones for (4.12) developed in [53], we find that two nodes $[\boldsymbol{\alpha}]^i$ and $[\boldsymbol{\beta}]^j$ are connected in the matrix setting if and only if $\boldsymbol{x}^{\boldsymbol{\alpha}}y_i$ and $\boldsymbol{x}^{\boldsymbol{\beta}}y_j$ are connected in the scalar setting. Moreover, the PMI sign symmetries of (4.1) are in a one-to-one correspondence to the usual sign symmetries of (4.12). Therefore, by [53, Corollary 6.8], the block structures of the term sparsity iterations with block closures for (4.1) converge to the one determined by the common PMI sign symmetries of $F(\boldsymbol{x}),G_1(\boldsymbol{x}),\ldots,G_m(\boldsymbol{x})$. ∎

Example 4.1. Consider the problem

$$\inf_{\boldsymbol{x}\in\mathbb{R}^2} \lambda_{\min}\left(F(\boldsymbol{x})=\begin{bmatrix} x_1^2 & x_1+x_2 \\ x_1+x_2 & x_2^2\end{bmatrix}\right) \quad \text{s.t.} \quad 1-x_1^2-x_2^2\geq 0. \tag{4.13}$$

Note that $F(\boldsymbol{x})$ has a nontrivial PMI sign symmetry $\theta=(-1,-1)$. With the relaxation order $r=2$, the procedure for exploiting term sparsity with block closures stabilizes at $s=2$, giving rise to two blocks for the moment matrix:

$$\left\{[1]^1,[x_1]^2,[x_2]^2,[x_1^2]^1,[x_1x_2]^1,[x_2^2]^1\right\}, \quad \left\{[1]^2,[x_1]^1,[x_2]^1,[x_1^2]^2,[x_1x_2]^2,[x_2^2]^2\right\}.$$

The corresponding sparse relaxation (4.1) yields a lower bound -0.9142 which coincides with the bound given by the second order dense relaxation.

5. Correlative sparsity. The aim of this section is to explore the correlative sparsity of the PMO problem (4.1):

$$\lambda^\star := \inf_{\boldsymbol{x}\in\mathbb{R}^n} \lambda_{\min}(F(\boldsymbol{x})) \quad \text{s.t.} \quad G_1(\boldsymbol{x})\succeq 0,\ldots,G_m(\boldsymbol{x})\succeq 0,$$

where $F\in\mathbb{S}^p[\boldsymbol{x}]$ and $G_k\in\mathbb{S}^{q_k}[\boldsymbol{x}],k\in[m]$.

We define the *correlative sparsity pattern (CSP) graph* of (4.1), denoted by $\mathcal{G}^{\text{csp}}$, such that $\mathcal{V}(\mathcal{G}^{\text{csp}})=[n]$ and $\{i,j\}\in\mathcal{E}(\mathcal{G}^{\text{csp}})$ if one of following conditions holds:

(i) There exists $\boldsymbol{\alpha}\in\text{supp}(F)$ such that $i,j\in\text{supp}(\boldsymbol{\alpha}):=\{k\in[n]\mid\alpha_k\neq 0\}$.

(ii) There exists $k\in[m]$ such that $x_i,x_j\in\ \text{var}(G_k)$, where $\text{var}(G_k)$ is the subset of variables effectively involved in G_k.

Let $\{\mathcal{I}_\ell\}_{\ell=1}^t$ be the list of maximal cliques of $\mathcal{G}^{\text{csp}}$ with $n_\ell:=|\mathcal{I}_\ell|$. For $\ell\in[t]$, let $\mathbb{R}[\boldsymbol{x}(\mathcal{I}_\ell)]$ denote the ring of polynomials in the n_ℓ variables $\boldsymbol{x}(\mathcal{I}_\ell):=\{x_i\mid i\in\mathcal{I}_\ell\}$. We can decompose the constraint polynomial matrices $G_1,\ldots,G_m$ into groups $\{G_k\mid k\in\mathcal{J}_\ell\},\ell=1,\ldots,t$, which satisfy

(i) $\cup_{\ell=1}^t\mathcal{J}_\ell=[m]$;

(ii) for any $k\in\mathcal{J}_\ell$, $\text{var}(G_k)\subseteq\mathcal{I}_\ell$, $\ell=1,\ldots,t$.

Given a sequence $\mathbf{S}=(S_{\boldsymbol{\alpha}})_{\boldsymbol{\alpha}\in\mathbb{N}^n}\subseteq\mathbb{S}^p$ and $\ell\in[t]$, for $d\in\mathbb{N}$ and $G\in\mathbb{R}[\boldsymbol{x}(\mathcal{I}_\ell)]$, let $M_d(\mathbf{S},\mathcal{I}_\ell)$ (resp., $M_d(G\mathbf{S},\mathcal{I}_\ell)$) be the moment (resp., localizing) submatrix obtained from $M_d(\mathbf{S})$ (resp., $M_d(G\mathbf{S})$) by retaining only those block rows and columns indexed by $\boldsymbol{\alpha}\in\mathbb{N}_d^n$ with $\text{supp}(\boldsymbol{\alpha})\subseteq\mathcal{I}_\ell$.

Then with $r\geq r_{\min}$, the rth order correlative sparsity adapted moment relaxation for (4.1) is given by

$$\lambda_r^{(\text{cs})} := \begin{cases} \inf & \mathscr{L}_{\mathbf{S}}(F) \\ \text{s.t.} & M_r(\mathbf{S},\mathcal{I}_\ell)\succeq 0, \quad \ell\in[t], \\ & M_{r-d_k}(G_k\mathbf{S},\mathcal{I}_\ell)\succeq 0, \quad k\in\mathcal{J}_\ell,\ell\in[t], \\ & \mathscr{L}_{\mathbf{S}}(I_p)=1, \end{cases} \tag{5.1}$$

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and its dual problem reads as

$$(5.2)\qquad \begin{cases} \sup & \lambda \\ \text{s.t.} & F-\lambda I_p=\sum_{\ell=1}^{t}\left(S_{\ell,0}+\sum_{k\in\mathcal{J}_\ell}\langle S_{\ell,k},G_k\rangle_p\right), \\ & S_{\ell,0}\in\Sigma^p[\boldsymbol{x}(\mathcal{I}_\ell)]_{2r},\ S_{\ell,k}\in\Sigma^{pq_k}[\boldsymbol{x}(\mathcal{I}_\ell)]_{2(r-d_k)}, \quad k\in\mathcal{J}_\ell,\ell\in[t]. \end{cases}$$

Each $\lambda_r^{(\mathrm{cs})}$ therefore provides a lower bound on $\lambda^\star$, and the sequence $\{\lambda_r^{(\mathrm{cs})}\}_{r\geq r_{\min}}$ is monotonically nondecreasing. Moreover, asymptotic convergence to $\lambda^\star$ can be guaranteed under appropriate conditions when $p=1$.

THEOREM 5.1 ([27, Theorem 1]). *Let $p=1$. Assume that the following conditions hold:*

(i) *The CSP $\{\mathcal{I}_\ell\}_{\ell=1}^t$ satisfies the running intersection property, i.e., for every $\ell\in[t-1]$, there exists $s\in[\ell]$ such that*

$$\mathcal{I}_{\ell+1}\cap\bigcup_{j=1}^{\ell}\mathcal{I}_j\subseteq\mathcal{I}_s.$$

(ii) *For each $\ell\in[t]$, there exists $a_\ell>0$ such that*

$$a_\ell-\sum_{i\in\mathcal{I}_\ell}x_i^2=\sigma_{\ell,0}+\sum_{k\in\mathcal{J}_\ell}\langle S_{\ell,k},G_k\rangle,$$

where $\sigma_{\ell,0}\in\Sigma^1[\boldsymbol{x}]$, $S_{\ell,k}\in\Sigma^{q_k}[\boldsymbol{x}]$.

Then $\lim_{r\to\infty}\lambda_r^{(\mathrm{cs})}=\lambda^\star$.

In view of Theorem 5.1, one may expect that the asymptotic convergence also holds true when $p>1$ under similar conditions. However, the following counterexample (inspired by [41]) shows that this is not the case even when $p=2$. In other words, the matrix counterpart (i.e., for $p>1$) of Putinar's Positivstellensatz for polynomials with correlative sparsity studied in [13, 27, 28] does not hold in general.

Example 5.1. Let

$$F(\boldsymbol{x})=\begin{bmatrix}2(x_1-1)^2+(x_2-1)^2+(x_2-2)^2 & 3-2x_2\\ 3-2x_2 & 2(x_1-2)^2+(x_2-1)^2+(x_2-2)^2\end{bmatrix}$$

and $\mathbf{K}=\left\{(x_1,x_2)\in\mathbb{R}^2:4-x_1^2\geq 0,\,4-x_2^2\geq 0\right\}$. Let

$$Q=\begin{bmatrix}\frac{13}{2} & 3 & -2 & -3 & -3 & 2\\ 3 & \frac{25}{2} & 3 & -4 & -4 & -3\\ -2 & 3 & 2 & 0 & 0 & -2\\ -3 & -4 & 0 & 2 & 2 & 0\\ -3 & -4 & 0 & 2 & 2 & 0\\ 2 & -3 & -2 & 0 & 0 & 2\end{bmatrix}.$$

The matrix Q has three eigenvalues, 0 and $\frac{27}{2}\pm 3\sqrt{2}$ (so $Q\succeq 0$), and

$$F(\boldsymbol{x})-\frac{1}{2}I_2=(\mathbf{m}_1(\boldsymbol{x})\otimes I_2)^\intercal Q(\mathbf{m}_1(\boldsymbol{x})\otimes I_2)\succeq 0.$$

Moreover, we have $(\frac{3}{2},1)\in\mathbf{K}$ and $\lambda_{\min}\left(F(\frac{3}{2},1)\right)=\frac{1}{2}$, so $\inf_{\boldsymbol{x}\in\mathbf{K}}\lambda_{\min}(F(\boldsymbol{x}))=\frac{1}{2}$ (whose minimizers form a circle: $(x_1-\frac{3}{2})^2+(x_2-\frac{3}{2})^2=\frac{1}{4}$). Hence, for any $\varepsilon>0$,

$$F(\boldsymbol{x})-\left(\frac{1}{2}-\varepsilon\right)I_2\succ 0\quad\forall\boldsymbol{x}\in\mathbf{K}.$$

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However, for any $\varepsilon \in (0, \frac{1}{2})$, we will show that the representation

$$F(\boldsymbol{x}) - \left(\frac{1}{2} - \varepsilon\right) I_2 = \sum_{i=1}^{2} \left(S_{i,0}(x_i) + (4 - x_i^2) S_{i,1}(x_i)\right), \tag{5.3}$$

where each $S_{i,j} \in \Sigma^2[x_i]$, does not hold. Suppose on the contrary that there exists a representation (5.3) for some $\varepsilon \in (0, \frac{1}{2})$. Let Φ_1, Φ_2 be the following matrix-valued atomic measures defined in the x_1-space and x_2-space, respectively:

$$\Phi_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \delta_{(x_1=1)} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \delta_{(x_1=2)}, \quad \Phi_2 = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \delta_{(x_2=1)} + \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \delta_{(x_2=2)}.$$

Then, we define a linear functional $\mathscr{L} : \mathbb{S}^2[x_1] + \mathbb{S}^2[x_2] \to \mathbb{R}$ by

$$\mathscr{L}\left(H_1(x_1) + H_2(x_2)\right) = \mathscr{L}_{\Phi_1}(H_1(x_1)) + \mathscr{L}_{\Phi_2}(H_2(x_2)) \quad \forall H_1(x_1) \in \mathbb{S}^2[x_1], H_2(x_2) \in \mathbb{S}^2[x_2].$$

Since $\mathscr{L}_{\Phi_1}(I_2) = \mathscr{L}_{\Phi_2}(I_2) = 1$, there is no ambiguity in the above definition. One can easily check that $\mathscr{L}(F) = 0$. Now applying $\mathscr{L}$ to both sides of (5.3), we obtain

$$0 - \left(\frac{1}{2} - \varepsilon\right) = \mathscr{L}\left(F(\boldsymbol{x}) - \left(\frac{1}{2} - \varepsilon\right) I_2\right) = \mathscr{L}\left(\sum_{i=1}^{2} \left(S_{i,0}(x_i) + (4 - x_i^2) S_{i,1}(x_i)\right)\right) \geq 0,$$

which is a contradiction.

Remark 5.1. We are not aware of any simple condition that can guarantee asymptotic convergence of the correlative sparsity adapted hierarchy for PMO with a matrix objective. This will be pursued in our future research. On the other hand, one may restore asymptotic convergence in the case of correlative sparsity by considering the scalarized problem $\inf_{\boldsymbol{x} \in \mathbf{K}, \|\boldsymbol{y}\|^2 = 1} \boldsymbol{y}^\intercal F(\boldsymbol{x}) \boldsymbol{y}$.

Even though asymptotic convergence does not hold for PMO with correlative sparsity in general, finite convergence may still occur in practice, and this convergence could be detected by flatness conditions. For the rest of this section, we establish several results on detecting finite convergence and extracting optimal solutions when exploiting correlative sparsity. For each $\ell \in [t]$, let us define

$$\mathbf{K}_\ell := \left\{ \boldsymbol{x}(\ell) \in \mathbb{R}^{|\mathcal{I}_\ell|} \,\middle|\, G_k(\boldsymbol{x}) \succeq 0 \quad \forall k \in \mathcal{J}_\ell \right\}, \tag{5.4}$$

where $\boldsymbol{x}(\ell) := (x_i)_{i \in \mathcal{I}_\ell}$.

THEOREM 5.2. *Let* $\mathbf{S}$ *be an optimal solution of* (5.1). *Assume that the following conditions hold:*

(i) *For each* $\ell \in [t]$, $\mathbf{S}^\ell := (S_{\boldsymbol{\alpha}})_{\mathrm{supp}(\boldsymbol{\alpha}) \subseteq \mathcal{I}_\ell}$ *admits a representing measure* $\Phi_\ell = S_{\mathbf{0}}(\sum_{i=1}^{s_\ell} \xi_{i\ell} \delta_{\boldsymbol{x}^i(\ell)})$ *for some* $\xi_{i\ell} > 0$ *with* $\sum_{i=1}^{s_\ell} \xi_{i\ell} = 1$ *and* $\boldsymbol{x}^1(\ell), \ldots, \boldsymbol{x}^{s_\ell}(\ell) \in \mathbf{K}_\ell$.

(ii) *For all pairs* $\{i, j\}$ *with* $\mathcal{I}_i \cap \mathcal{I}_j \neq \emptyset$, *one has* $\mathrm{rank}(M_r(\mathbf{S}, \mathcal{I}_i \cap \mathcal{I}_j)) = 1$.

Then $\lambda_r^{(\mathrm{cs})} = \lambda^\star$. *Moreover, each point* $\hat{\boldsymbol{x}} \in \mathbb{R}^n$ *satisfying* $\hat{\boldsymbol{x}}(\ell) := (\hat{x}_i)_{i \in \mathcal{I}_\ell} = \boldsymbol{x}^{i_\ell}(\ell)$ *for some* $i_\ell \in [s_\ell]$, $\ell \in [t]$, *is an optimal solution of* (4.1).

Proof. For each $\ell \in [t]$, let us pick a point $\boldsymbol{x}^{i_\ell}(\ell)$, $i_\ell \in [s_\ell]$, and then define the point $\hat{\boldsymbol{x}} \in \mathbb{R}^n$ such that $\hat{\boldsymbol{x}}(\ell) := (\hat{x})_{i \in \mathcal{I}_\ell} = \boldsymbol{x}^{i_\ell}(\ell)$, $\ell \in [t]$. Because $\mathrm{rank}(M_r(\mathbf{S}, \mathcal{I}_i \cap \mathcal{I}_j)) = 1$ for all pairs $\{i, j\}$ with $\mathcal{I}_i \cap \mathcal{I}_j \neq \emptyset$, the value of $\hat{x}_k$ is unique for any $k \in \mathcal{I}_i \cap \mathcal{I}_j \neq \emptyset$. Therefore, $\hat{\boldsymbol{x}}$ is well defined and $\hat{\boldsymbol{x}} \in \mathbf{K}$. We can thus construct $s = \prod_{j=1}^{\ell} s_\ell$ solutions

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$\{\boldsymbol{x}^j\}_{j=1}^s \subseteq \mathbf{K}$, each associated with the weight matrix $W_j := \prod_{\ell=1}^t \xi_{i_\ell \ell} S_{\mathbf{0}} \in \mathbb{S}_+^p$ if $\hat{\boldsymbol{x}}(\ell) = \boldsymbol{x}^{i_\ell}(\ell)$ for some $i_\ell \in [s_\ell]$. We then define the following matrix measure $\Phi = \sum_{j=1}^s W_j \delta_{\boldsymbol{x}^j}$ which is supported on $\mathbf{K}$, and its marginal measure on $\mathbf{K}_\ell$ is Φ_ℓ for each $\ell \in [t]$. Moreover, it holds that

$$\sum_{j=1}^s W_j = \prod_{\ell=1}^t \left(\sum_{i=1}^{s_\ell} \xi_{i\ell} \right) S_{\mathbf{0}} = S_{\mathbf{0}}. \tag{5.5}$$

Therefore, we have

$$\begin{aligned} \lambda^\star \geq \lambda_r^{(\mathrm{cs})} = \mathscr{L}_{\mathbf{S}}(F) &= \sum_{j=1}^s \langle W_j, F(\boldsymbol{x}^j) \rangle \\ &\geq \sum_{j=1}^s \langle W_j, \lambda^\star I_p \rangle = \lambda^\star \left\langle \sum_{j=1}^s W_j, I_p \right\rangle = \lambda^\star \mathrm{tr}\,(S_{\mathbf{0}}) = \lambda^\star, \end{aligned}$$

which implies $\lambda_r^{(\mathrm{cs})} = \lambda^\star$ and each $\boldsymbol{x}^j$ is an optimal solution of (4.1). ∎

Example 5.2. Consider the PMO problem

$$\begin{cases} \inf\limits_{\boldsymbol{x} \in \mathbb{R}^3} & \lambda_{\min}\left((-x_1^2 + x_2)(\boldsymbol{q}_1 \boldsymbol{q}_1^\intercal + \boldsymbol{q}_2 \boldsymbol{q}_2^\intercal) + (x_2^2 + x_3^2) \boldsymbol{q}_3 \boldsymbol{q}_3^\intercal \right) \\ \text{s.t.} & 1 - x_1^2 - x_2^2 \geq 0,\ x_2^2 + x_3^2 = 1, \end{cases} \tag{5.6}$$

where $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ are the column vectors of the orthogonal matrix

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{bmatrix}. \tag{5.7}$$

Problem (5.6) exhibits a CSP: $\mathcal{I}_1 = \{1, 2\}$ and $\mathcal{I}_2 = \{2, 3\}$. Solving (5.1) with $r = 2$, we obtain $\lambda_2^{(\mathrm{cs})} = -1.250$ and $\mathrm{rank}(M_2(\mathbf{S}, \mathcal{I}_1)) = \mathrm{rank}(M_1(\mathbf{S}, \mathcal{I}_1)) = 4$, $\mathrm{rank}(M_2(\mathbf{S}, \mathcal{I}_2)) = \mathrm{rank}(M_1(\mathbf{S}, \mathcal{I}_2)) = 4$. Applying the extraction procedure in [15] to $M_2(\mathbf{S}, \mathcal{I}_1)$ and $M_2(\mathbf{S}, \mathcal{I}_2)$, respectively, we retrieve the matrix measures:

$$\Phi_1 \approx W\left(\delta_{\boldsymbol{x}^1(1)} + \delta_{\boldsymbol{x}^2(1)}\right) \text{ and } \Phi_2 \approx W\left(\delta_{\boldsymbol{x}^1(2)} + \delta_{\boldsymbol{x}^2(2)}\right) \tag{5.8}$$

with

$$W \approx \begin{bmatrix} 0.2083 & -0.0833 & 0.0416 \\ -0.0833 & 0.0833 & 0.0833 \\ 0.0416 & 0.0833 & 0.2083 \end{bmatrix} \tag{5.9}$$

and

$$\boldsymbol{x}^1(1) \approx \begin{bmatrix} 0.8660 \\ -0.4999 \end{bmatrix}, \boldsymbol{x}^2(1) \approx \begin{bmatrix} -0.8660 \\ -0.4999 \end{bmatrix}, \boldsymbol{x}^1(2) \approx \begin{bmatrix} -0.4999 \\ 0.8660 \end{bmatrix}, \boldsymbol{x}^2(2) \approx \begin{bmatrix} -0.4999 \\ -0.8660 \end{bmatrix}.$$

We can merge Φ_1 and Φ_2 into $\Phi \approx W\left(\delta_{\hat{\boldsymbol{x}}^1} + \delta_{\hat{\boldsymbol{x}}^2} + \delta_{\hat{\boldsymbol{x}}^3} + \delta_{\hat{\boldsymbol{x}}^4}\right)$ with

$$\hat{\boldsymbol{x}}^1 \approx \begin{bmatrix} 0.8660 \\ -0.4999 \\ 0.8660 \end{bmatrix}, \hat{\boldsymbol{x}}^2 \approx \begin{bmatrix} 0.8660 \\ -0.4999 \\ -0.8660 \end{bmatrix}, \hat{\boldsymbol{x}}^3 \approx \begin{bmatrix} -0.8660 \\ -0.4999 \\ 0.8660 \end{bmatrix}, \hat{\boldsymbol{x}}^4 \approx \begin{bmatrix} -0.8660 \\ -0.4999 \\ -0.8660 \end{bmatrix}. \tag{5.10}$$

Thus, by Theorem (5.2), the optimum of (5.6) is -1.250 which is achieved at $\hat{\boldsymbol{x}}^1$, $\hat{\boldsymbol{x}}^2$, $\hat{\boldsymbol{x}}^3$, and $\hat{\boldsymbol{x}}^4$.

For each $\ell \in [t]$, let $d_{\mathbf{K}_\ell} := \max_{j \in \mathcal{J}_\ell} \lceil \deg G_j / 2 \rceil$.

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COROLLARY 5.3. *Let* $\mathbf{S}$ *be an optimal solution of* (5.1). *Assume that the following conditions hold:*

(i) $\operatorname{rank}(S_{\mathbf{0}}) = 1$.

(ii) *For each* $\ell \in [t]$, *one has*

$$\operatorname{rank}(M_r(\mathbf{S}, \mathcal{I}_\ell)) = \operatorname{rank}(M_{r-d_{\mathbf{K}_\ell}}(\mathbf{S}, \mathcal{I}_\ell))(=: s_\ell). \tag{5.11}$$

(iii) *For all pairs* $\{i, j\}$ *with* $\mathcal{I}_i \cap \mathcal{I}_j \neq \emptyset$, *one has* $\operatorname{rank}(M_r(\mathbf{S}, \mathcal{I}_i \cap \mathcal{I}_j)) = 1$.

Then $\lambda_r^{(\mathrm{cs})} = \lambda^\star$. *Moreover, let* $\Delta_\ell := \{\boldsymbol{x}^i(\ell)\}_{i=1}^{s_\ell}$ *be the set of points obtained by applying the extraction procedure in* [15] *to each moment matrix* $M_r(\mathbf{S}, \mathcal{I}_\ell)$, $\ell \in [t]$, *and let* $S_{\mathbf{0}} = \boldsymbol{v}\boldsymbol{v}^\intercal$ *for some* $\boldsymbol{v} \in \mathbb{R}^p$. *Then each point* $\hat{\boldsymbol{x}} \in \mathbb{R}^n$ *satisfying* $\hat{\boldsymbol{x}}(\ell) := (\hat{x}_i)_{i \in \mathcal{I}_\ell} = \boldsymbol{x}^{i_\ell}(\ell)$ *for some* $\boldsymbol{x}^{i_\ell}(\ell) \in \Delta_\ell$, $\ell \in [t]$, *is an optimal solution of* (4.1) *and* $\boldsymbol{v}$ *is the corresponding eigenvector.*

Proof. By (5.11) and Theorem 2.1, for each $\ell \in [t]$, the sequence of matrices $\mathbf{S}^\ell := (S_{\boldsymbol{\alpha}})_{|\boldsymbol{\alpha}| \leq 2r, \operatorname{supp}(\boldsymbol{\alpha}) \subseteq \mathcal{I}_\ell}$ admits an s_ℓ-atomic matrix measure Φ_ℓ supported on $\mathbf{K}_\ell$, $\ell \in [t]$, so that

$$\Phi_\ell = \sum_{i=1}^{s_\ell} W_{i\ell} \delta_{\boldsymbol{x}^i(\ell)} \text{ for some } W_{i\ell} \in \mathbb{S}_+^p. \tag{5.12}$$

Note that $\sum_{i=1}^{s_\ell} W_{i\ell} = S_{\mathbf{0}}$ for each $\ell \in [t]$. The fact that $\operatorname{rank}(S_{\mathbf{0}}) = 1$ together with $W_{i\ell} \succeq 0$ implies that $\operatorname{rank}(W_{i\ell}) = 1$ for all i, ℓ, from which we deduce that there exists $\xi_{i\ell} > 0$ such that $W_{i\ell} = \xi_{i\ell} S_{\mathbf{0}}$ for all i, ℓ and $\sum_{i=1}^{s_\ell} \xi_{i\ell} = 1, \ell \in [t]$. Then the conclusion follows from Theorem 5.2. ∎

THEOREM 5.4. *Let* $\mathbf{S}$ *be an optimal solution of* (5.1). *Assume that the following conditions hold:*

(i) *There exist weight matrices* $W_1, \ldots, W_s \in \mathbb{S}_+^p$ *such that for each* $\ell \in [t]$, $\mathbf{S}^\ell := (S_{\boldsymbol{\alpha}})_{\operatorname{supp}(\boldsymbol{\alpha}) \subseteq \mathcal{I}_\ell}$ *admits a representing matrix measure* $\Phi_\ell = \sum_{i=1}^s W_i \delta_{\boldsymbol{x}^i(\ell)}$ *for some points* $\boldsymbol{x}^1(\ell), \ldots, \boldsymbol{x}^s(\ell) \subseteq \mathbf{K}_\ell$.

(ii) *For each* $i \in [s]$, *the points* $\boldsymbol{x}^i(1), \ldots, \boldsymbol{x}^i(t)$ *can be merged into a single point* $\hat{\boldsymbol{x}}^i \in \mathbb{R}^n$ *such that* $\hat{\boldsymbol{x}}^i(\ell) = \boldsymbol{x}^i(\ell)$ *for every* $\ell \in [t]$.

Then $\lambda_r^{(\mathrm{cs})} = \lambda^\star$. *Moreover, each point* $\hat{\boldsymbol{x}}^i, i \in [s]$ *is an optimal solution of* (4.1).

Proof. Let us define the matrix measure $\Phi = \sum_{i=1}^s W_i \delta_{\hat{\boldsymbol{x}}^i}$ which is supported on $\mathbf{K}$, and its marginal measure on $\mathbf{K}_\ell$ is Φ_ℓ for each $\ell \in [t]$. Then,

$$\begin{aligned}
\lambda^\star \geq \lambda_r^{(\mathrm{cs})} = \mathscr{L}_{\mathbf{S}}(F) &= \sum_{i=1}^s \left\langle W_i, F(\hat{\boldsymbol{x}}^i) \right\rangle \\
&\geq \sum_{i=1}^s \langle W_i, \lambda^\star I_p \rangle = \lambda^\star \left\langle \sum_{i=1}^s W_i, I_p \right\rangle = \lambda^\star \operatorname{tr}(S_{\mathbf{0}}) = \lambda^\star,
\end{aligned}$$

which implies $\lambda_r^{(\mathrm{cs})} = \lambda^\star$ and each $\hat{\boldsymbol{x}}^i$ is an optimal solution of (4.1). ∎

Example 5.3. Consider the PMO problem

$$\inf_{\boldsymbol{x} \in \mathbb{R}^3} \lambda_{\min} \left(\begin{bmatrix} x_1^2 + x_2^2 & 2 + x_1 x_2 + x_3^2 \\ 2 + x_1 x_2 + x_3^2 & x_2 x_3 \end{bmatrix} \right) \text{ s.t. } 1 - x_1^2 - x_2^2 \geq 0,\ 1 - x_2^2 - x_3^2 \geq 0, \tag{5.13}$$

which exhibits a CSP: $\mathcal{I}_1 = \{1, 2\}$ and $\mathcal{I}_2 = \{2, 3\}$. Solving (5.1) with $r = 2$, we obtain $\lambda_2^{(\mathrm{cs})} \approx -3.0643$ and $\operatorname{rank}(M_2(\mathbf{S}, \mathcal{I}_1)) = \operatorname{rank}(M_1(\mathbf{S}, \mathcal{I}_1)) = 2$, $\operatorname{rank}(M_2(\mathbf{S}, \mathcal{I}_2)) =$

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$\operatorname{rank}(M_1(\mathbf{S},\mathcal{I}_2)) = 2$. Applying the extraction procedure in [15] to $M_2(\mathbf{S},\mathcal{I}_1)$ and $M_2(\mathbf{S},\mathcal{I}_2)$, respectively, we retrieve the matrix measures

$$\Phi_1 \approx \begin{bmatrix} 0.2338 & -0.2494 \\ -0.2494 & 0.2661 \end{bmatrix} \delta_{\boldsymbol{x}^1(1)} + \begin{bmatrix} 0.2338 & -0.2494 \\ -0.2494 & 0.2661 \end{bmatrix} \delta_{\boldsymbol{x}^2(1)}, \tag{5.14}$$

$$\Phi_2 \approx \begin{bmatrix} 0.2338 & -0.2494 \\ -0.2494 & 0.2661 \end{bmatrix} \delta_{\boldsymbol{x}^1(2)} + \begin{bmatrix} 0.2338 & -0.2494 \\ -0.2494 & 0.2661 \end{bmatrix} \delta_{\boldsymbol{x}^2(2)} \tag{5.15}$$

with

$$\boldsymbol{x}^1(1) \approx \begin{bmatrix} 0.2732 \\ 0.2561 \end{bmatrix}, \boldsymbol{x}^2(1) \approx \begin{bmatrix} -0.2732 \\ -0.2561 \end{bmatrix}, \boldsymbol{x}^1(2) \approx \begin{bmatrix} 0.2561 \\ -0.9666 \end{bmatrix}, \boldsymbol{x}^2(2) \approx \begin{bmatrix} -0.2561 \\ 0.9666 \end{bmatrix}.$$

We can merge Φ_1 and Φ_2 into

$$\Phi \approx \begin{bmatrix} 0.2338 & -0.2494 \\ -0.2494 & 0.2661 \end{bmatrix} \delta_{\hat{\boldsymbol{x}}^1} + \begin{bmatrix} 0.2338 & -0.2494 \\ -0.2494 & 0.2661 \end{bmatrix} \delta_{\hat{\boldsymbol{x}}^2} \tag{5.16}$$

with

$$\hat{\boldsymbol{x}}^1 \approx \begin{bmatrix} 0.2732 \\ 0.2561 \\ -0.9666 \end{bmatrix}, \quad \hat{\boldsymbol{x}}^2 \approx \begin{bmatrix} -0.2732 \\ -0.2561 \\ 0.9666 \end{bmatrix}. \tag{5.17}$$

Thus, by Theorem (5.4), the optimum of (5.13) is -3.0643 which is achieved at $\hat{\boldsymbol{x}}^1$ and $\hat{\boldsymbol{x}}^2$.

COROLLARY 5.5. *Let* $\mathbf{S}$ *be an optimal solution of* (5.1). *If* $\operatorname{rank}(M_{r_{\min}}(\mathbf{S},\mathcal{I}_\ell)) = 1$ *for each* $\ell \in [t]$, *then* $\lambda_r^{(\mathrm{cs})} = \lambda^\star$. *Moreover, one can recover an optimal solution of* (4.1).

Proof. Since $\operatorname{rank}(M_{r_{\min}}(\mathbf{S},\mathcal{I}_\ell)) = 1$, by [15, Theorem 5], each sequence $\mathbf{S}^\ell := (S_{\boldsymbol{\alpha}})_{|\boldsymbol{\alpha}|\le 2r_{\min},\operatorname{supp}(\boldsymbol{\alpha})\subseteq\mathcal{I}_\ell}$ admits a Dirac representing measure $\Phi_\ell = S_{\mathbf{0}}\delta_{\boldsymbol{x}(\ell)}$ with $\boldsymbol{x}(\ell) \in \mathbf{K}_\ell$. Moreover, we can merge $\boldsymbol{x}(1),\ldots,\boldsymbol{x}(t)$ into a point $\hat{\boldsymbol{x}} \in \mathbb{R}^n$ by letting $\hat{\boldsymbol{x}}(\mathcal{I}_\ell) := (\hat{x}_i)_{i\in\mathcal{I}_\ell} = \boldsymbol{x}(\ell) \quad \forall \ell \in [t]$. There is no ambiguity for $\hat{x}_i$ when $i \in \mathcal{I}_j \cap \mathcal{I}_k \neq \emptyset$ for some $j,k \in [t]$. In fact, denoting by $\boldsymbol{e}_i \in \mathbb{N}^n$ the vector whose ith entry is 1 and the others are 0, we have $S_{\boldsymbol{e}_i} = S_{\mathbf{0}}\boldsymbol{x}(j)_i = S_{\mathbf{0}}\boldsymbol{x}(k)_i$, which implies $\boldsymbol{x}(j)_i = \boldsymbol{x}(k)_i$. Then, the conclusion follows from Theorem 5.4. ∎

6. Matrix sparsity. In this section, we briefly introduce how to exploit matrix sparsity (i.e., chordal sparsity pattern of objective or constraint polynomial matrices) to perform reductions on the SDP size for PMO (4.1). The cases for objective polynomial matrices and for constraint polynomial matrices will be addressed sequentially.

6.1. Objective matrix sparsity. To exploit chordal sparsity encoded in the objective matrix, we adapt Theorem 2.4, Theorem 2.5, and Corollary 2.2 in [55] dealing with scalar polynomial constraints to the case of PMI constraints. For the proofs, one only needs to replace the usual inner product $\langle\cdot,\cdot\rangle$ in the corresponding proofs in [55] with the product $\langle\cdot,\cdot\rangle_p$. We thus omit the details.

THEOREM 6.1. *Let* $\mathbf{K}$ *be the semialgebraic set defined in* (2.5), *and suppose that* $\Sigma^p[\mathbf{G}]$ *is Archimedean. Let* $F(\boldsymbol{x})$ *be a polynomial matrix whose sparsity graph is chordal and has maximal cliques* $\mathcal{C}_1,\ldots,\mathcal{C}_t$. *If* $F(\boldsymbol{x})$ *is strictly positive definite on* $\mathbf{K}$, *then there exist SOS matrices* $S_{k,i}(\boldsymbol{x})$ *of size* $q_k|\mathcal{C}_i| \times q_k|\mathcal{C}_i|$ *such that*

$$F(\boldsymbol{x}) = \sum_{i=1}^{t} E_{\mathcal{C}_i}^{\intercal}\left(S_{0,i}(\boldsymbol{x}) + \sum_{k=1}^{m}\langle S_{k,i}(\boldsymbol{x}), G_k(\boldsymbol{x})\rangle_p\right)E_{\mathcal{C}_i}.$$

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THEOREM 6.2. *Let $\mathbf{K}$ be a semialgebraic set defined in* (2.5) *with homogeneous polynomial matrices $G_1, \ldots, G_m$ of even degree and such that $\mathbf{K} \setminus \{\mathbf{0}\}$ is nonempty. Let $F(\boldsymbol{x})$ be a homogeneous polynomial matrix of even degree whose sparsity graph is chordal and has maximal cliques $\mathcal{C}_1, \ldots, \mathcal{C}_t$. If $F(\boldsymbol{x})$ is strictly positive definite on $\mathbf{K} \setminus \{\mathbf{0}\}$, then there exist an integer $\tau \geq 0$ and homogeneous SOS matrices $S_{k,i}(\boldsymbol{x})$ of size $q_k|\mathcal{C}_i| \times q_k|\mathcal{C}_i|$ such that*

$$\|\boldsymbol{x}\|^{2\tau} F(\boldsymbol{x}) = \sum_{i=1}^{t} E_{\mathcal{C}_i}^{\intercal} \left(S_{0,i}(\boldsymbol{x}) + \sum_{k=1}^{m} \langle S_{k,i}(\boldsymbol{x}), G_k(\boldsymbol{x}) \rangle_p \right) E_{\mathcal{C}_i}.$$

COROLLARY 6.3. *Let $\mathbf{K}$ be a semialgebraic set defined in* (2.5)*, and let $F(\boldsymbol{x})$ be an inhomogeneous polynomial matrix of even degree whose sparsity graph is chordal and has maximal cliques $\mathcal{C}_1, \ldots, \mathcal{C}_t$. If $F(\boldsymbol{x})$ is strictly positive definite on $\mathbf{K}$ and its highest-degree homogeneous part is strictly positive definite on $\mathbb{R}^n \setminus \{\mathbf{0}\}$, then there exist an integer $\tau \geq 0$ and SOS matrices $S_{k,i}(\boldsymbol{x})$ of size $q_k|\mathcal{C}_i| \times q_k|\mathcal{C}_i|$ such that*

$$(1 + \|\boldsymbol{x}\|)^{2\tau} F(\boldsymbol{x}) = \sum_{i=1}^{t} E_{\mathcal{C}_i}^{\intercal} \left(S_{0,i}(\boldsymbol{x}) + \sum_{k=1}^{m} \langle S_{k,i}(\boldsymbol{x}), G_k(\boldsymbol{x}) \rangle_p \right) E_{\mathcal{C}_i}.$$

Then, the decomposed optimization problem for (4.1) based on Theorems 6.1, 6.2, and Corollary 6.3 reads as

$$\sup \lambda \quad \text{s.t.} \quad F(\boldsymbol{x}) - \lambda = \sum_{i=1}^{t} E_{\mathcal{C}_i}^{\intercal} \left(S_{0,i}(\boldsymbol{x}) + \sum_{k=1}^{m} \langle S_{k,i}(\boldsymbol{x}), G_k(\boldsymbol{x}) \rangle_p \right) E_{\mathcal{C}_i},$$

$$\sup \lambda \quad \text{s.t.} \quad \|\boldsymbol{x}\|^{2\tau}(F(\boldsymbol{x}) - \lambda) = \sum_{i=1}^{t} E_{\mathcal{C}_i}^{\intercal} \left(S_{0,i}(\boldsymbol{x}) + \sum_{k=1}^{m} \langle S_{k,i}(\boldsymbol{x}), G_k(\boldsymbol{x}) \rangle_p \right) E_{\mathcal{C}_i},$$

$$\sup \lambda \quad \text{s.t.} \quad (1 + \|\boldsymbol{x}\|)^{2\tau}(F(\boldsymbol{x}) - \lambda) = \sum_{i=1}^{t} E_{\mathcal{C}_i}^{\intercal} \left(S_{0,i}(\boldsymbol{x}) + \sum_{k=1}^{m} \langle S_{k,i}(\boldsymbol{x}), G_k(\boldsymbol{x}) \rangle_p \right) E_{\mathcal{C}_i},$$

respectively.

6.2. Constraint matrix sparsity. We now consider the case that the constraint matrix $G(\boldsymbol{x})$ has a chordal sparsity graph $\mathcal{G}$. One way to exploit constraint matrix sparsity is decomposing a single PMI constraint into multiple ones of smaller sizes according to its sparsity pattern (by Theorem 2.2), which most often brings additional correlative sparsity to exploit. Let us illustrate this method by a concrete example.

Example 6.1. Consider the PMO problem

$$\inf_{\boldsymbol{x} \in \mathbb{R}^5} \lambda_{\min}(F(\boldsymbol{x})) \quad \text{s.t.} \quad G(\boldsymbol{x}) \succeq 0 \tag{6.1}$$

with

$$F(\boldsymbol{x}) = \begin{bmatrix} x_1 & x_2 & x_3 \\ x_2 & 1 & x_4 \\ x_3 & x_4 & x_5 \end{bmatrix} \text{ and } G(\boldsymbol{x}) = \begin{bmatrix} 1 - x_1^2 - x_2^2 - x_3^2 & x_1x_2x_3 & 0 \\ x_1x_2x_3 & x_3 & x_3x_4x_5 \\ 0 & x_3x_4x_5 & 1 - x_3^2 - x_4^2 - x_5^2 \end{bmatrix}.$$

By introducing an auxiliary variable x_6 and Theorem 2.2, we can decompose the PMI constraint $G(\boldsymbol{x}) \succeq 0$ as

$$G_1(x_1, x_2, x_3, x_6) = \begin{bmatrix} 1 - x_1^2 - x_2^2 - x_3^2 & x_1x_2x_3 \\ x_1x_2x_3 & x_6^2 \end{bmatrix} \succeq 0,$$

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$$G_2(x_3,x_4,x_5,x_6)=\begin{bmatrix} x_3-x_6^2 & x_3x_4x_5 \\ x_3x_4x_5 & 1-x_3^2-x_4^2-x_5^2 \end{bmatrix}\succeq 0$$

such that $G\succeq 0$ if and only if $G_1\succeq 0, G_2\succeq 0$. Then (6.1) becomes

$$\inf_{\boldsymbol{x}\in\mathbb{R}^6}\lambda_{\min}(F(\boldsymbol{x}))\quad \text{s.t.}\quad G_1(x_1,x_2,x_3,x_6)\succeq 0,\quad G_2(x_3,x_4,x_5,x_6)\succeq 0, \tag{6.2}$$

which exhibits a CSP: $\mathcal{I}_1=\{1,2,3,6\}$ and $\mathcal{I}_2=\{3,4,5,6\}$.

In the appendix of the preprint version of the current paper,[2] we provide an alternative way to exploit constraint matrix sparsity.

7. Numerical examples. The sparsity-adapted relaxations for PMO problems have been implemented in the Julia package `TSSOS`[3] [32]. In this section, we provide numerical examples to illustrate the efficiency of our approach with `TSSOS` 1.4.4, where `Mosek` 11.0 [3] is employed as an SDP solver with default settings. When presenting the results in tables, the column labelled "mb" records the maximal PSD block size of SDP relaxations, the column labelled "bound" records lower bounds given by SDP relaxations, and the column labelled "time" records computational time in seconds. Moreover, r stands for the relaxation order, s stands for the sparse order, and the symbol "-" indicates that `Mosek` runs out of memory. For brevity, TS denotes term sparsity, CS denotes correlative sparsity, and MS denotes matrix sparsity. All numerical experiments were performed on a desktop computer with an Intel(R) Core(TM) i9-14900 CPU@2.00 GHz and 128G RAM.[4]

7.1. Term sparse examples. Our first example involves the objective matrix

$$F(\boldsymbol{x})=\begin{bmatrix} x_1^4 & x_1^2-x_2x_3 & x_3^2-x_4x_5 & x_1x_4 & x_1x_5 \\ x_1^2-x_2x_3 & x_2^4 & x_2^2-x_3x_4 & x_2x_4 & x_2x_5 \\ x_3^2-x_4x_5 & x_2^2-x_3x_4 & x_3^4 & x_4^2-x_1x_2 & x_5^2-x_3x_5 \\ x_1x_4 & x_2x_4 & x_4^2-x_1x_2 & x_4^4 & x_4^2-x_1x_3 \\ x_1x_5 & x_2x_5 & x_5^2-x_3x_5 & x_4^2-x_1x_3 & x_5^4 \end{bmatrix} \tag{7.1}$$

and constraint matrices

$$G_1(\boldsymbol{x})=\begin{bmatrix} 1-x_1^2-x_2^2 & x_2x_3 \\ x_2x_3 & 1-x_3^2 \end{bmatrix},\quad G_2(\boldsymbol{x})=\begin{bmatrix} 1-x_4^2 & x_4x_5 \\ x_4x_5 & 1-x_5^2 \end{bmatrix}. \tag{7.2}$$

The considered optimization problem aims to minimize the minimum eigenvalue of F over the region defined by G_1,G_2:

$$\inf_{\boldsymbol{x}\in\mathbb{R}^5}\lambda_{\min}(F(\boldsymbol{x}))\quad \text{s.t.}\quad G_1(\boldsymbol{x})\succeq 0,\, G_2(\boldsymbol{x})\succeq 0. \tag{7.3}$$

Table 1 reports the numerical results of this example, where we compare the dense approach with the sparse approach (exploiting term sparsity with block/chordal closures). It can be seen that except for the case of performing chordal closures with $s=2$,[5] the sparse approach is several times faster than the dense approach while yielding the same lower bound -2.2766. The bound -2.2766 is certified as the global optimum by checking the flatness condition. For this example, the term sparsity iteration with block closures stabilizes at $s=2$.

[2]It is available at https://arxiv.org/abs/2411.15479.

[3]`TSSOS` is freely available at https://github.com/wangjie212/TSSOS.

[4]The script for reproducing the results is available at https://github.com/wangjie212/TSSOS/blob/master/example/pmi.jl.

[5]The chordal closure yields PSD blocks with overlaps. When such overlaps are significant, the computation may get slow.

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TABLE 1
Results for problem (7.3).

	$r=2$			$r=3$			$r=4$		
	mb	bound	time	mb	bound	time	mb	bound	time
Dense	105	-2.2766	0.31	280	-2.2766	5.34	630	-2.2766	99.7
TS (block, $s=1$)	50	-2.2766	0.14	132	-2.2766	1.97	330	-2.2766	30.3
TS (block, $s=2$)	80	-2.2766	0.15	200	-2.2766	2.39	430	-2.2766	35.7
TS (chordal, $s=1$)	48	-2.2766	0.18	58	-2.2766	0.51	58	-2.2766	0.93
TS (chordal, $s=2$)	80	-2.2766	0.29	200	-2.2766	8.05	430	-2.2766	225

TABLE 2
Results for problem (7.4) *with* $r=2$.

	$p=20$			$p=40$			$p=60$		
	mb	bound	time	mb	bound	time	mb	bound	time
Dense	200	-20.259	6.13	400	-39.388	185	600	-59.560	1756
TS (block, $s=1$)	80	-20.259	0.37	160	-39.388	7.76	240	-59.560	55.4
TS (block, $s=2$)	80	-20.259	0.33	160	-39.388	7.05	240	-59.560	51.7
TS (chordal, $s=1$)	60	-20.259	0.36	120	-39.388	8.41	180	-59.560	68.1
TS (chordal, $s=2$)	80	-20.259	0.37	160	-39.388	12.6	240	-59.560	55.1

Next for a positive integer $p \in \{20, 40, 60\}$, we consider the problem

$$\inf_{\boldsymbol{x}\in\mathbb{R}^3} \lambda_{\min}\left((1-x_1^2-x_2^2)I_p + (x_1^2-x_3^2)A + (x_1^2x_3^2-2x_2^2)B\right) \text{ s.t. } G(\boldsymbol{x}) \succeq 0 \tag{7.4}$$

with

$$G(\boldsymbol{x}) = \begin{bmatrix} 1-x_1^2-x_2^2 & x_2x_3 \\ x_2x_3 & 1-x_3^2 \end{bmatrix}. \tag{7.5}$$

The matrices $A, B \in \mathbb{S}^p$ are generated with entries being drawn randomly from the uniform distribution on $(0,1)$. Table 2 reports the numerical results of this problem, where we compare the dense approach with the sparse approach (exploiting term sparsity with block or chordal closures). It can be seen that the sparse approach is significantly faster and more scalable than the dense approach while yielding the same lower bounds. For this problem, the term sparsity iteration with block closures stabilizes at $s=2$.

7.2. Correlatively sparse examples. Consider the problem

$$\inf_{\boldsymbol{x}\in\mathbb{R}^5} \lambda_{\min}(F(\boldsymbol{x})) \quad \text{s.t.} \quad G_1(\boldsymbol{x}) \succeq 0, G_2(\boldsymbol{x}) \succeq 0 \tag{7.6}$$

with

$$F(\boldsymbol{x}) = \begin{bmatrix} x_1^4 & x_1^2-x_2x_3 & x_3^2-x_4x_5 & 0.5 & 0.5 \\ x_1^2-x_2x_3 & x_2^4 & x_2^2-x_3x_4 & 0.5 & 0.5 \\ x_3^2-x_4x_5 & x_2^2-x_3x_4 & x_3^4 & x_4^2-x_1x_2 & x_5^2-x_3x_4 \\ 0.5 & 0.5 & x_4^2-x_1x_2 & x_4^4 & x_4^2-x_1x_3 \\ 0.5 & 0.5 & x_5^2-x_3x_4 & x_4^2-x_1x_3 & x_5^4 \end{bmatrix} \tag{7.7}$$

and

$$G_1(\boldsymbol{x}) = \begin{bmatrix} 1-x_1^2-x_2^2 & x_2x_3 \\ x_2x_3 & 1-x_3^2 \end{bmatrix}, \quad G_2(\boldsymbol{x}) = \begin{bmatrix} 1-x_4^2 & x_4x_5 \\ x_4x_5 & 1-x_5^2 \end{bmatrix}, \tag{7.8}$$

which exhibits a CSP: $\mathcal{I}_1 = \{1,2,3\}$, $\mathcal{I}_2 = \{3,4\}$, and $\mathcal{I}_3 = \{4,5\}$.

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TABLE 3
Results for problem (7.6).

r	Dense			CS			CS + Block			CS + Chordal		
	mb	bound	time	mb	bound	time	mb	bound	time	mb	bound	time
2	105	-2.4131	0.31	50	-2.4131	0.06	26	-2.4131	0.03	12	-2.4148	0.02
3	280	-2.4131	5.27	100	-2.4131	0.26	62	-2.4131	0.10	16	-2.4131	0.04
4	630	-2.4131	121	200	-2.4131	1.23	74	-2.4131	0.23	16	-2.4131	0.08

TABLE 4
Results for problem (7.11) *with* $r = 2$.

n	Dense			CS			CS + Block			CS + Chordal		
	mb	bound	time	mb	bound	time	mb	bound	time	mb	bound	time
5	63	-1.0247	0.09	30	-1.0247	0.03	16	-1.1931	0.02	5	-1.5918	0.02
7	108	-1.1157	0.38	30	-1.1157	0.07	16	-1.3072	0.03	5	-1.9648	0.03
9	165	-1.3891	2.42	30	-1.3891	0.07	16	-1.4522	0.04	5	-2.3454	0.05
11	234	-1.7620	9.06	30	-1.7620	0.09	16	-1.7620	0.05	5	-2.7288	0.06
13	315	-2.1403	38.2	30	-2.1403	0.11	16	-2.1403	0.05	5	-3.1136	0.08

Table 3 reports the numerical results of this example, where we compare the dense approach with three sparse approaches: exploiting correlative sparsity, exploiting both correlative sparsity and term sparsity with block or chordal closures. It could be seen that the approach exploiting correlative sparsity runs much faster than the dense approach without sacrificing the accuracy, and taking term sparsity into account brings some extra speedup.

The next example involves a PMO problem with an increasing number of variables. For a given integer $n \geq 3$, let

$$F(\boldsymbol{x}) = \begin{bmatrix} \sum_{k=1}^{n-2} x_k^2 & \sum_{k=1}^{n-1} x_k x_{k+1} & 1 \\ \sum_{k=1}^{n-1} x_k x_{k+1} & \sum_{k=2}^{n-1} x_k^2 & \sum_{k=1}^{n-2} x_k x_{k+2} \\ 1 & \sum_{k=1}^{n-2} x_k x_{k+2} & \sum_{k=3}^{n} x_k^2 \end{bmatrix} \tag{7.9}$$

and

$$G_k(\boldsymbol{x}) = \begin{bmatrix} 1 - x_k^2 - x_{k+1}^2 & x_{k+1} + 0.5 \\ x_{k+1} + 0.5 & 1 - x_{k+2}^2 \end{bmatrix}, \quad k \in [n-2], \tag{7.10}$$

and consider the optimization problem

$$\inf_{\boldsymbol{x} \in \mathbb{R}^n} \lambda_{\min}(F(\boldsymbol{x})) \quad \text{s.t.} \quad G_k(\boldsymbol{x}) \succeq 0,\, k \in [n-2]. \tag{7.11}$$

Problem (7.11) possesses a CSP: $\mathcal{I}_k = \{k, k+1, k+2\}, k \in [n-2]$. Table 4 reports the related numerical results with $r = 2$ as n increases. As for the previous example, we compare the dense approach with three sparse approaches: exploiting correlative sparsity, exploiting both correlative sparsity and term sparsity with block or chordal closures. We could see that the approach exploiting correlative sparsity yields the same bounds as the dense approach but in much less time (especially when the number of variables n is large). Taking term sparsity into account could further reduce computational time but provides looser bounds.

7.3. Matrix sparse examples. In this subsection, we solve three PMO problems with matrix sparsity. The first problem is

$$\inf_{\boldsymbol{x} \in \mathbb{R}^5} \lambda_{\min}(F(\boldsymbol{x})) \quad \text{s.t.} \quad G_1(\boldsymbol{x}) \succeq 0, G_2(\boldsymbol{x}) \succeq 0 \tag{7.12}$$

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TABLE 5
Results for problem (7.12).

r	Dense			MS			MS + Block			MS + Chordal		
	mb	bound	time	mb	bound	time	mb	bound	time	mb	bound	time
2	105	-2.4180	0.28	63	-2.4180	0.11	33	-2.4180	0.06	18	-2.4180	0.05
3	280	-2.4180	5.35	168	-2.4180	1.71	90	-2.4180	0.60	27	-2.4180	0.17
4	630	-2.4180	118	378	-2.4180	23.8	198	-2.4180	8.54	63	-2.4180	0.83

with

$$F(\boldsymbol{x}) = \begin{bmatrix} x_1^4 & x_1^2 - x_2x_3 & x_3^2 - x_4x_5 & & \\ x_1^2 - x_2x_3 & x_2^4 & x_2^2 - x_3x_4 & & \\ x_3^2 - x_4x_5 & x_2^2 - x_3x_4 & x_3^4 & x_4^2 - x_1x_2 & x_5^2 - x_3x_4 \\ & & x_4^2 - x_1x_2 & x_4^4 & x_4^2 - x_1x_3 \\ & & x_5^2 - x_3x_4 & x_4^2 - x_1x_3 & x_5^4 \end{bmatrix} \tag{7.13}$$

and

$$G_1(\boldsymbol{x}) = \begin{bmatrix} 1 - x_1^2 - x_2^2 & x_2x_3 \\ x_2x_3 & 1 - x_3^2 \end{bmatrix}, \quad G_2(\boldsymbol{x}) = \begin{bmatrix} 1 - x_4^2 & x_4x_5 \\ x_4x_5 & 1 - x_5^2 \end{bmatrix}. \tag{7.14}$$

Table 5 reports the related numerical results of problem (7.12), where we compare the dense approach with three sparse approaches: exploiting matrix sparsity, exploiting both matrix sparsity and term sparsity with block or chordal closures. We could see that the three sparse approaches all yield the same bound as the dense approach but in much less time, and exploiting term sparsity brings further speedup.

Now we consider

$$\inf_{\boldsymbol{x} \in \mathbb{R}^5} \lambda_{\min}\left(\begin{bmatrix} x_1^4 + x_2^4 + 1 & x_1x_3 \\ x_1x_3 & x_3^4 + x_4^4 + x_5^4 + 0.5 \end{bmatrix}\right) \quad \text{s.t.} \quad G(\boldsymbol{x}) \succeq 0 \tag{7.15}$$

with

$$G(\boldsymbol{x}) = \begin{bmatrix} 1 - x_1^2 & x_1x_2 & x_1x_3 & & \\ x_1x_2 & 1 - x_2^2 & x_2x_3 & & \\ x_1x_3 & x_2x_3 & 1 - x_3^2 & x_3x_4 & x_3x_5 \\ & & x_3x_4 & 1 - x_4^2 & x_4x_5 \\ & & x_3x_5 & x_4x_5 & 1 - x_5^2 \end{bmatrix}. \tag{7.16}$$

By introducing a new variable x_6, the PMI constraint $G(\boldsymbol{x}) \succeq 0$ can be decomposed as

$$G_1(x_1, x_2, x_3, x_6) = \begin{bmatrix} 1 - x_1^2 & x_1x_2 & x_1x_3 \\ x_1x_2 & 1 - x_2^2 & x_2x_3 \\ x_1x_3 & x_2x_3 & x_6^2 \end{bmatrix} \succeq 0,$$

$$G_2(x_3, x_4, x_5, x_6) = \begin{bmatrix} 1 - x_3^2 - x_6^2 & x_3x_4 & x_3x_5 \\ x_3x_4 & 1 - x_4^2 & x_4x_5 \\ x_3x_5 & x_4x_5 & 1 - x_5^2 \end{bmatrix} \succeq 0$$

such that $G \succeq 0$ if and only if $G_1 \succeq 0, G_2 \succeq 0$. Note that after the decomposition, the PMO problem exhibits a CSP: $\mathcal{I}_1 = \{1, 2, 3, 6\}, \mathcal{I}_2 = \{3, 4, 5, 6\}$.

Table 6 reports the related numerical results of problem (7.15), where we compare the dense approach with three sparse approaches: exploiting both matrix sparsity and

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TABLE 6
Results for problem (7.15).

r	Dense			MS + CS			MS + CS + Block			MS + CS + Chordal		
	mb	bound	time	mb	bound	time	mb	bound	time	mb	bound	time
2	60	0.3977	0.05	30	0.3977	0.04	6	0.3977	0.01	5	0.3977	0.01
3	210	0.3977	0.82	90	0.3977	0.17	14	0.3977	0.02	7	0.3977	0.02
4	560	0.3977	4.94	210	0.3977	1.55	26	0.3977	0.05	15	0.3977	0.05

correlative sparsity, exploiting matrix sparsity, correlative sparsity, and term sparsity with block or chordal closures. It could be seen that the three sparse approaches all yield the same bound as the dense approach while in much less time (especially when $r \geq 3$), and exploiting term sparsity brings extra speedup.

We next consider a PMO problem adapted from [25, sect. 6]:

$$\inf_{\boldsymbol{x}\in\mathbb{R}^n} \lambda_{\min}\left(\begin{bmatrix} 1 & x_1x_2 \\ x_1x_2 & 1+x_n^2 \end{bmatrix}\right) \quad \text{s.t.} \quad G(\boldsymbol{x}) \succeq 0 \tag{7.17}$$

with

$$G(\boldsymbol{x}) = \begin{bmatrix} 1-x_1^4 & 0 & 0 & \cdots & 0 & x_1x_2 \\ 0 & 1-x_2^4 & 0 & \cdots & 0 & x_2x_3 \\ 0 & 0 & 1-x_3^4 & \cdots & 0 & x_3x_4 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1-x_{n-1}^4 & x_{n-1}x_n \\ x_1x_2 & x_2x_3 & x_3x_4 & \cdots & x_{n-1}x_n & 1-x_n^4 \end{bmatrix} \succeq 0. \tag{7.18}$$

By introducing $n-2$ new variables $\{x_{n+1}, \ldots, x_{2n-2}\}$, the PMI constraint $G(\boldsymbol{x}) \succeq 0$ can be decomposed as

$$G_1(x_1, x_2, x_{n+1}) = \begin{bmatrix} 1-x_1^4 & x_1x_2 \\ x_1x_2 & x_{n+1}^2 \end{bmatrix} \succeq 0,$$
$$G_{n-1}(x_{n-1}, x_n, x_{2n-2}) = \begin{bmatrix} 1-x_{n-1}^4 & x_{n-1}x_n \\ x_{n-1}x_n & 1-x_n^4-x_{2n-2}^2 \end{bmatrix} \succeq 0,$$

and

$$G_i(x_i, x_{i+1}, x_{n+i-1}, x_{n+i}) = \begin{bmatrix} 1-x_i^4 & x_ix_{i+1} \\ x_ix_{i+1} & x_{n+i}^2 - x_{n+i-1}^2 \end{bmatrix} \succeq 0, \quad i = 2, \ldots, n-2,$$

such that $G \succeq 0$ if and only if $G_i \succeq 0, i \in [n-1]$. Note that after the decomposition, the PMO problem exhibits a CSP: $\mathcal{I}_1 = \{1, 2, n+1\}, \mathcal{I}_i = \{i, i+1, n+i-1, n+i\} (i = 2, \ldots, n-2), \mathcal{I}_{n-1} = \{n-1, n, 2n-2\}$.

Table 7 reports the numerical results of problem (7.17) with $n \in \{5, 7, 9, 11, 13\}$, where we compare the dense approach with three sparse approaches: exploiting both matrix sparsity and correlative sparsity, exploiting matrix sparsity, correlative sparsity, and term sparsity with block or chordal closures. One could see that the three sparse approaches all yield the same bound as the dense approach while in much less time when $n \in \{5, 7\}$, and scale much better with large n.

7.4. Randomly generated sparse examples. Finally, we test randomly generated sparse examples. Given $t \in \mathbb{N} \setminus \{0\}$, let $\mathcal{I}_i = \{3i-2, \ldots, 3i+2\}$ and let $F_i(\boldsymbol{x}(\mathcal{I}_i)), G_i(\boldsymbol{x}(\mathcal{I}_i)) \in \mathbb{S}^4[\boldsymbol{x}(\mathcal{I}_i)]$ be random polynomial matrices whose entries are

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TABLE 7
Results for problem (7.17) *with* $r = 4$.

n	Dense			MS + CS			MS + CS + Block			MS + CS + Chordal		
	mb	bound	time	mb	bound	time	mb	bound	time	mb	bound	time
5	252	0.2138	2.81	140	0.2138	0.80	15	0.2138	0.03	15	0.2138	0.03
7	660	0.2138	138	140	0.2138	1.23	15	0.2138	0.05	15	0.2138	0.05
9	1430	-	-	140	0.2138	2.15	15	0.2138	0.07	15	0.2138	0.07
11	2730	-	-	140	0.2138	2.64	15	0.2138	0.09	15	0.2138	0.09
13	4760	-	-	140	0.2138	3.24	15	0.2138	0.10	15	0.2138	0.10

TABLE 8
Results for problem (7.19) *with* $r = 2$.

t	Dense			CS			CS + block			CS + chordal		
	mb	bound	time	mb	bound	time	mb	bound	time	mb	bound	time
2	180	-2.4648	2.86	96	-2.4648	0.41	27	-2.5195	0.07	7	-2.5722	0.06
4	480	-3.5626	365	96	-3.7565	0.89	80	-3.8478	0.41	14	-4.1282	0.20
6	924	-	-	96	-4.8454	1.50	80	-4.9889	0.57	8	-5.5581	0.19
8	1512	-	-	96	-6.5244	2.07	80	-6.5449	0.88	14	-7.0377	0.40
10	2244	-	-	96	-7.8778	3.18	80	-7.8858	1.16	13	-8.2506	0.46

sparse quadratic polynomials with coefficients randomly chosen according to the standard normal distribution for $i = 1, \ldots, t$. Then we consider the following PMO problem:

$$
\begin{cases}
\inf\limits_{\boldsymbol{x} \in \mathbb{R}^{3t+2}} & \lambda_{\min}\left(\sum_{i=1}^{t} F_i(\boldsymbol{x}(\mathcal{I}_i))\right) \\
\text{s.t.} & G_i(\boldsymbol{x}(\mathcal{I}_i)) + 2\sum_{j \in \mathcal{I}_i} x_j^2 \cdot I_4 \succeq 0, \quad i = 1, \ldots, t, \\
& 1 - \sum_{j \in \mathcal{I}_i} x_j^2 \geq 0, \quad i = 1, \ldots, t.
\end{cases}
\tag{7.19}
$$

Table 8 reports the numerical results of problem (7.19) with $t \in \{2, 4, 6, 8, 10\}$, where we compare the dense approach with three sparse approaches: exploiting correlative sparsity, exploiting both correlative sparsity and term sparsity with block or chordal closures. One could see that the three sparse approaches run much faster and scale significantly better than the dense approach while providing slightly weaker bounds.

8. Conclusions. This paper has explored the use of various sparsity methods in reducing the size of matrix Moment-SOS relaxations for verification of PMIs and solving PMO problems. We have shown that multiple sparsity structures, e.g., term sparsity, correlative sparsity, and matrix sparsity, could be simultaneously exploited to maximize the dimensionality reduction of the derived SDP in the PMO setting. These methods make the matrix Moment-SOS hierarchy more capable of tackling practical applications, as illustrated by extensive numerical experiments in which structured PMO problems can be tractably solved, whereas the dense hierarchy most often runs out of memory. The sparsity routines introduced in this paper were incorporated into the open-source `TSSOS` package and are thus available to interested practitioners in fields such as optimization, control, and operations research.

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